Pure Math 450 - Lebesgue Integration and Fourier Analysis

Lecture 1: Inner Product Spaces and Hilbert Spaces

Definition 1.1
Let X be a vector space over $\mathbb{R}$ (it is also common to use $\mathbb{C}$). An inner product on X is a function $\langle \cdot, \cdot \rangle : X \times X \to \mathbb{R}$ satisfying,
(1)
$$\langle \alpha_1 x_1 + \alpha_2 x_2, y \rangle = \alpha_1 \langle x_1, y \rangle + \alpha_2 \langle x_2, y \rangle$$
for all $x_1, x_2, y \in X$, $\alpha_1, \alpha_2 \in \mathbb{R}$ (Bilinearity in the first argument).
(2)
$$\langle x, \beta_1 y_1 + \beta_2 y_2 \rangle = \beta_1 \langle x, y_1 \rangle + \beta_2 \langle x, y_2 \rangle$$
for all $x, y_1, y_2 \in X$, $\beta_1, \beta_2 \in \mathbb{R}$ (Bilinearity in the second argument).
(3)
$$\langle x, y \rangle = \langle y, x \rangle$$
for all $x, y \in X$ (symmetry).
(4)
$$\langle x, x \rangle > 0$$
for all $x \in X$ such that $x \neq 0_X$ (positive definiteness).
In this case, the pair $(X, \langle \cdot, \cdot \rangle)$ is called an inner product space (over $\mathbb{R}$).

Remark 1.2
(1) One sees that symmetry and bilinearity in the first argument implies bilinearity in the second argument, so there is some redundancy in our definition.
(2) By bilinearity and symmetry, we have, for all $y \in X$,
$$\langle 0_X, y \rangle = \langle y, 0_X \rangle = \langle y, 20_X \rangle = 2\langle y, 0_X \rangle$$
and so $\langle 0_X, y \rangle = 2\langle 0_X, y \rangle$, so $\langle 0_X, y \rangle = 0$. Therefore another way to phrase positive definiteness is that $\langle x, x \rangle \geq 0$ for all $x \in X$, with equality if and only if $x = 0_X$.

Notation 1.3
Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. For every $x \in X$, we denote
$$||x|| := \sqrt{\langle x, x \rangle} \in [0, \infty)$$
which we call the norm of x.

Proposition 1.4
Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space.
(1 - Cauchy-Schwarz inequality) For all $x, y \in X$,
$$|\langle x, y \rangle| \leq ||x|| \; ||y||$$
with equality if and only if x and y are linearly dependent.
(2) The norm of Notation 1.3 is a norm on X.
Proof:
The proof of (1) is an exercise, which has been seen before (for example, in linear algebra).
To verify that $|| \cdot ||$ is a norm, the fact that $||x|| \geq 0$ for all $x \in X$ with equality if and only if $x = 0_X$ follows from positive definiteness of $\langle \cdot, dt \rangle$, and $||\alpha x|| = |\alpha| \; ||x||$ for all $\alpha \in \mathbb{R}, x \in X$ follows from bilinearity of the inner product. To verify the triangle inequality, we have, for all $x, y \in X$,
$$||x+y||^2 = \langle x+y, x+y \rangle \overbrace{=}^{\text{bilin.}} \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle = ||x||^2 + 2\langle x, y \rangle + ||y||^2$$

$$\leq ||x||^2 + 2|\langle x, y\rangle| + ||y||^2 \overbrace{\leq}^{\text{CS}} ||x||^2 + 2||x|| \; ||y|| + ||y||^2 = (||x|| + ||y||)^2$$

and taking square roots gives the result. ■

Definition 1.5 (Hilbert Space)
Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. In light of Proposition 1.4, we can view $(X, || \cdot ||)$ as a normed vector space, and thus a metric space (X, d) with distance $d(x, y) = ||x - y||$ for $x, y \in X$. If X is complete with respect to the distance d, or in other words, if $(X, || \cdot ||)$ is a Banach space, then we say that $(X, \langle \cdot, \cdot \rangle)$ is a Hilbert space.

Example 1.6
Take $X = \mathbb{R}^k$, with $\langle \cdot, \cdot \rangle$ the standard inner product given by

$$\langle x, y\rangle = \sum_{i=1}^{k} x^{(i)} y^{(i)}$$

for $x = (x^{(1)}, \ldots, x^{(k)}), y = (y^{(1)}, \ldots, y^{(k)}) \in \mathbb{R}^k$ is a Hilbert space, as seen previously in calculus. In this case,

$$||x|| = \sqrt{[x^{(1)}]^2 + \cdots + [x^{(k)}]^2}$$

Example 1.7
Consider the space c_{00}, the space of eventually zero real sequences, i.e.

$$c_{00} = \{(x^{(1)}, \ldots, x^{(k)}, \ldots) : x^{(k)} \in \mathbb{R} \; \forall k \in \mathbb{N}, \exists \; N \in \mathbb{N} \; s.t. \; x^{(k)} = 0 \; \forall k \geq N \in \mathbb{N}\}$$

This has an inner product

$$\langle x, y\rangle = \sum_{k=1}^{\infty} x^{(k)} y^{(k)}$$

for $x = (x^{(1)}, \ldots, x^{(k)}, \ldots), y = (y^{(1)}, \ldots, y^{(k)}, \ldots) \in c_{00}$, but it is not a Hilbert space, since it is not complete. Indeed, its induced norm is

$$||x||_2 = \sqrt{\sum_{k=1}^{\infty} x^{(k)^2}}$$

As an exercise, give an example of a Cauchy sequence in $(c_{00}, || \cdot ||_2)$ which is not convergent. So how do we get a Hilbert space? All we need to do is complete the space, that is, embed c_{00} into a larger Banach space where c_{00} is dense. The completion here is ℓ^2, the space of square-summable sequences, that is,

$$\ell^2 = \left\{ x = (x^{(1)}, \ldots, x^{(n)}, \ldots) : \sum_{i=1}^{\infty} x^{(i)} < \infty \right\}$$

Some facts about ℓ^2:
(1) For $x, y \in \ell^2$, $\sum_{i=1}^{\infty} x^{(i)} y^{(i)}$ is absolutely convergent. We can then define $\langle x, y\rangle = \sum_{i=1}^{\infty} x^{(i)} y^{(i)}$.
(2) With the above inner product, ℓ^2 is an inner product space.
(3) $(\ell^2, || \cdot ||_2)$ is complete, so $(\ell^2, \langle \cdot, \cdot \rangle)$ is a Hilbert space.
(4) c_{00} is dense in $(\ell^2, || \cdot ||_2)$.
Some of these claims will be on assignment 1.

Example 1.8
Pick $a < b \in \mathbb{R}$, and consider the space $M = \mathcal{C}([a, b], \mathbb{R})$, the space of real-valued continuous functions on

$[a, b]$, with pointwise addition and scalar multiplication to make it into a vector space. In PMath 351, we saw $(M, ||\cdot||_\infty)$ is a Banach space. We can define another norm on this space via an inner product:

$$\langle f, g\rangle = \int_0^1 f(t)g(t)dt$$

Checking this isn't an inner product isn't very hard, the hardest part is showing $\langle f, f\rangle = 0$ if and only if $f = 0$, which requires the continuity of f (argue by contradiction). However, we have the same issue as above, where this isn't a Hilbert space (use continuous approximations of a step function), how do we complete this space? The space we are looking for is called $L^2([a, b])$, which will show up later in the course.

Lecture 2: Some Hilbert space geometry: distance to a closed convex set

Definition 2.1
Let X be a vector space (over $\mathbb{R}$).
(1) For $x, y \in X$ define the line segment with endpoints x and y (or the convex hull of x and y) as $\mathrm{Co}(x, y) = \{(1-t)x + ty : t \in [0, 1]\}$, where an expression of the form $(1-t)x + ty$ for $t \in [0, 1]$ is called a convex combination of x and y.
(2) A set $A \subseteq X$ is called convex if for all $x, y \in A$, $\mathrm{Co}(x, y) \subseteq A$, that is, $(1-t)x + ty \in A$ for all $t \in [0, 1], x, y \in A$.

Remark 2.2
(1) Let $(X, ||\cdot||)$ be a normed vector space with $A \subseteq X$. If A is convex, then $\mathrm{cl}(A)$, its closure (with respect to $||\cdot||$), is convex as well. The proof of this fact was a homework question in PMath 351. To see this, use the description of the closure in terms of convergent sequences.
(2) Let (X, d) be a metric space and let $A \subseteq X$ be a nonempty closed set. As seen in PMath 351, one defines a "distance to A" function,

$$d_A : X \to \mathbb{R}$$

given by $d_A(x) = \inf\{d(x, a) : a \in A\}$. This satisfies the following properties:
- $d_A(x) \geq 0$ for all $x \in X$ with $d_A(x) = 0$ if and only if $x \in A$ (this uses the fact A is closed).
- $d_A(x)$ is continuous, in fact, it is 1-Lipschitz: $|d_A(x_1) - d_A(x_2)| \leq d(x_1, x_2)$ for all $x_1, x_2 \in X$.

Theorem 2.3
Let $(X, \langle\cdot,\cdot\rangle)$ be a Hilbert space. Suppose $A \subseteq X$ is a non-empty closed convex set, and let $x_0 \in X$. Then there exists a unique $a_0 \in A$ such that $||x_0 - a_0|| = d_A(x_0)$.
Proof:
In the case when $x_0 \in A$, taking $a_0 = x_0$ suffices, since then $d_A(x_0) = 0$ and x_0 is the only point in X that satisfies $d(x_0, x_0) = ||x_0 - x_0|| = 0$. Thus, assume $x_0 \notin A$. Let $\alpha := d_A(x_0) > 0$.
To prove uniqueness, assume for a contradiction there exist distinct points $a_1, a_2 \in A$ such that $||x_0 - a_1|| = ||x_0 - a_2|| = \alpha$. Let $\beta := ||a_1 - a_2|| > 0$. Let $a_3 = \frac{1}{2}(a_1 + a_2)$, which is a convex combination of a_1 and a_2, so $a_3 \in A$ by convexity. Q5(c) on homework 1 says that

$$||x_0 - a_3|| \overbrace{=}^{\text{Q5(c)}} \sqrt{\alpha^2 - \beta^2/4} < \alpha$$

where the last inequality is because $\beta > 0$. So then $||x_0 - a_3|| < \alpha = \inf\{||x_0 - a|| : a \in A\}$, which is a contradiction since $a_3 \in A$.
To prove existence, since $\alpha = \inf\{||x_0 - a|| : a \in A\}$, by definition of the infimum, we have

$$||x_0 - a|| \geq \alpha \quad (\diamond)$$

for all $a \in A$. Furthermore, for all $n \in \mathbb{N}$, we can find a point $a_n \in A$ such that

$$||x_0 - a_n|| < \alpha + \frac{1}{n} \quad (\diamond\diamond)$$

This gives a sequence $(a_n)_{n=1}^{\infty}$ of points in A. We claim that for every $m, n \in \mathbb{N}$, we have

$$||a_m - a_n||^2 \leq 4\alpha \left(\frac{1}{m} + \frac{1}{n}\right) + \left(\frac{2}{m^2} + \frac{2}{n^2}\right) \leq 4\alpha \left(\frac{1}{m} + \frac{1}{n}\right) + \left(\frac{2}{m} + \frac{2}{n}\right) = (4\alpha + 2)\left(\frac{1}{m} + \frac{1}{n}\right)$$

To see this, write

$$||a_m - a_n||^2 = ||(x_0 - a_n) - (x_0 - a_m)||^2 \overbrace{=}^{\text{Q4(a), HW1}} 2||x_0 - a_m||^2 + 2||x_0 - a_n||^2 - ||(x_0 - a_n) + (x_0 - a_m)||^2$$

$$\overbrace{\leq}^{(\diamond\diamond)} 2(\alpha + \frac{1}{m})^2 + 2(\alpha + \frac{1}{n})^2 - ||(x_0 - a_n) + (x_0 - a_m)||^2$$

For the third term, we have

$$||(x_0 - a_n) + (x_0 - a_m)||^2 = ||2x_0 - (a_n + a_m)||^2 = ||2(x_0 - \frac{1}{2}(a_m + a_n))||^2 = 4||x_0 - \frac{1}{2}(a_m + a_n)||^2$$

$$\overbrace{\geq}^{(\diamond)} 4\alpha^2$$

where $(\diamond)$ holds since $1/2(a_m + a_n) \in A$ by convexity. Combining this with the above estimate, we have

$$||a_m - a_n||^2 \leq 2(\alpha + \frac{1}{m})^2 + 2(\alpha + \frac{1}{n})^2 - 4\alpha^2 = 4\alpha \left(\frac{1}{m} + \frac{1}{n}\right) + \left(\frac{2}{m^2} + \frac{2}{n^2}\right)$$

as claimed. With this, we see $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence. Indeed, let $\varepsilon > 0$. Pick $n_0 \in \mathbb{N}$ large enough such that $\frac{8\alpha+4}{n_0} < \varepsilon^2$. Then for every $m, n \geq n_0$, we have

$$||a_m - a_n||^2 \leq (4\alpha + 2)(\frac{1}{m} + \frac{1}{n}) \leq (4\alpha + 2)(\frac{2}{n_0}) = \frac{8\alpha + 4}{n_0} < \varepsilon^2$$

and so taking square roots we get that $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence.
Therefore, by completeness of X, $(a_n)_{n=1}^{\infty}$ converges, so there exists $a_0 := \lim_{n\to\infty} a_n$, which is in A by closure of A. We claim this satisfies $||x_0 - a_0|| = \alpha$, which will complete the proof. Indeed, for every $n \in \mathbb{N}$, we have

$$\alpha \overbrace{\leq}^{(\diamond)} ||x_0 - a_0|| \leq ||x_0 - a_n|| + ||a_n - a_0|| \overbrace{\leq}^{(\diamond\diamond)} \alpha + \frac{1}{n} + ||a_n - a_0|| \overset{n\to\infty}{\to} \alpha$$

and so by the squeeze theorem, we get $||x_0 - a_0|| = \alpha$. ■

Remark 2.4
In Theorem 2.3, the completeness of X is an essential hypothesis to prove existence, but we did not use it to prove uniqueness. To see it is essential, let $X = c_{oo}$, which is not a Hilbert space. One can find a nonempty closed convex set $A \subseteq c_{oo}$ and a point $x_0 \notin A$ such that for every point $a \in A$, we have $||x_0 - a|| > d_A(x_0)$.

Lecture 3: Orthogonal projection onto a closed linear subspace of a Hilbert space

Remark 3.1
(1) Let X be a vector space, and $W \subseteq X$ a linear subspace. Then we see that W is a nonempty (since it contains 0) convex set.
(2) Now let $(X, ||\cdot||)$ be a normed vector space and $W \subseteq X$ a linear subspace that is closed with respect to $||\cdot||$. Then this means $W \subseteq X$ is an example of a nonempty closed convex set.

Definition 3.2
Let $(X, \langle\cdot,\cdot\rangle)$ be an inner product space. We say $x, y \in X$ are orthogonal to each other if $\langle x, y\rangle = 0$, and denote this by $x \perp y$. We say $A, B \subseteq X$ are orthogonal to each other if $x \perp y$ for all $x \in A, y \in B$. Note that $x \perp x$ means $\langle x, x\rangle = 0$, implying $x = 0$.

Theorem 3.3
Let $(X, \langle\cdot,\cdot\rangle)$ be a Hilbert space, and let $W \subseteq X$ be a closed linear subspace, with $x_0 \in X$. Let $w_0 \in W$ be the unique point such that $||x_0 - w_0|| = d_W(x_0)$ (courtesy of Theorem 2.3). Then $(x_0 - w_0) \perp W$.
Proof:
We must prove $\langle x_0 - w_0, w\rangle = 0$ for all $w \in W$. Suppose for the sake of contradiction there exists $w_1 \in W$ such that $\langle x_0 - w_0, w_1\rangle \neq 0$. Replacing w_1 by $-w_1$ if necessary, we may without loss of generality assume that $\langle x_0 - w_0, w_1\rangle > 0$. Now, for all $t \in \mathbb{R}$, we have $w_0 + tw_1 \in W$, and thus

$$||x_0 - (w_0 + tw_1)|| \geq d_W(x_0) \quad (\star)$$

Now for all $t \in \mathbb{R}$, observe

$$||x_0 - (w_0 + tw_1)||^2 = ||(x_0 - w_0) - tw_1||^2 = \langle (x_0 - w_0) - tw_1, (x_0 - w_0) - tw_1\rangle$$

$$= \langle x_0 - w_0, x_0 - w_0\rangle - 2\langle x_0 - w_0, tw_1\rangle + t^2\langle w_1, w_1\rangle = ||x_0 - w_0||^2 - 2t\langle x_0 - w_0, w_1\rangle + t^2||w_1||^2$$

$$= d_W(x_0)^2 - t||w_1||^2\left(\frac{2\langle x_0 - w_0, w_1\rangle}{||w_1||^2} - t\right)$$

Let $\delta = \frac{2\langle x_0 - w_0, w_1\rangle}{||w_1||^2}$, which is positive by the assumption $\langle x_0 - w_0, w_1\rangle > 0$. Now, if we pick $t = \delta/2$, combining this with $(\star)$, we get

$$d_W(x_0)^2 \underbrace{\leq}_{(\star)} ||x_0 - (w_0 + \frac{\delta}{2}w_1)||^2 = d_W(x_0)^2 - \underbrace{\frac{\delta^2||w_1||^2}{4}}_{>0} < d_W(x_0)^2$$

which is a contradiction. ∎

Remark 3.3'
Consider the above setup, where we now get a point $w_0 \in W$ such that $(x_0 - w_0) \perp W$. Then it follows that

$$||x_0 - w|| > ||x_0 - w_0||$$

for all $w \neq w_0 \in W$. Indeed, we have

$$||x_0 - w||^2 = ||(x_0 - w_0) + (w_0 - w)||^2 = ||x_0 - w_0||^2 + ||w_0 - w||^2 > ||x_0 - w_0||^2$$

since $(x_0 - w_0) \perp (w_0 - w) \in W$ (so we can apply Pythagoras) and $w \neq w_0$ so $||w_0 - w||^2 > 0$. Taking square roots, we get the claim. Therefore, w_0 satisfies the minimal distance condition of Theorem 2.3, so w_0 is the unique point such that $||x_0 - w_0|| = d_W(x_0)$.

Notation 3.3"
Let $(X, \langle\cdot,\cdot\rangle)$ be a Hilbert space, and $W \subseteq X$ a closed linear subspace. For every $x_0 \in X$, we denote $P_W(x_0)$ to be the unique point $w_0 \in W$ which is at minimal distance from x_0. $P_W(x_0)$ is called the orthogonal projection of x_0 onto W. The name is justified by Theorem 3.3, which says that $(x_0 - P_W(x_0)) \perp W$. It is customary to view P_W as a function from X to X (even though its image is always W), that is, a linear operator on X. Furthermore, it is contractive.

Definition/Remark 3.4
Let $(X, \langle\cdot,\cdot\rangle)$ be a Hilbert space, and let W be a closed linear subspace of X. The orthogonal complement of W is

$$W^\perp := \{x \in X : \langle x, w\rangle = 0, \forall w \in W\}$$

Then, $W^\perp$ is a closed linear subspace of X. To see this, it is convenient to write

$$W^\perp = \bigcap_{w\in W} \{x \in X : \langle x, w\rangle = 0\} = \bigcap_{w\in W} Y_w$$

where for $w \in W$ we put $Y_w := \{x \in X : \langle x, w\rangle = 0\}$. In order to check that $W^\perp$ is a closed linear subspace of X, it suffices to check that every Y_w is a closed linear subspace. To check closedness, one can use the sequential definition of a closed space, using Cauchy-Schwarz.

Exercise 3.5
(1) $W = \{0_X\}$ implies $W^\perp = X$. Indeed, we get

$$W^\perp = \{x \in X : x \perp 0_X\} = X$$

(2) $W = X$ implies $W^\perp = \{0_X\}$. Indeed, we get

$$W^\perp = \{x \in X : x \perp y, \forall y \in X\}$$

In particular, any $x \in W^\perp$ satisfies $x \perp x$, which forces $x = 0_X$, so $W^\perp = \{0_X\}$.
(3) For any closed linear subspace, $W \subseteq X$, we have $W \cap W^\perp = \{0_X\}$. The reverse inclusion is clear since W and $W^\perp$ are subspaces. For the forward inclusion, if $x \in W \cap W^\perp$, then $x \perp x$, forcing $x = 0_X$.

Exercise 3.6
Let $(X, \langle\cdot,\cdot\rangle)$ be a Hilbert space. Let $W \subseteq X$ be a closed linear subspace. Consider the function $P_W : X \to X$.
(1) Prove that P_W is linear.
(2) Prove that P_W is a contraction, that is, $||P_W(x)|| \leq ||x||$ for all $x \in X$.
(3) Prove that $\mathrm{im}(P_W) = W$.
To prove this, (1) and (2) are homework 2, question 4. To prove (3), the forward inclusion is true since $P_W(x) \in W$ for all $x \in X$, so $\mathrm{im}(P_W) \subseteq W$. For the reverse inclusion, we have $P_W(x) = x$ for all $x \in W$ since $x - x = 0_X \perp W$ for all $x \in W$, so by Remark 3.3' we have $P_W(x) = x$, and so $W \subseteq \mathrm{im}(PW)$. ■

Proposition 3.7
Let $(X, \langle\cdot,\cdot\rangle)$ be a Hilbert space and $W \subseteq X$ a closed linear subspace. Then every $x \in X$ can be written as $x = w + y$, where $w \in W, y \in W^\perp$. Moreover this decomposition is unique.
Proof:
For existence, we can write $x = P_W(x) + (x - P_W(x))$, where $P_W(x) \in W$ and $x - P_W(x) \in W^\perp$ by Theorem 3.3. Thus take $w = P_W(x)$ and take $y = x - P_W(x)$.
For uniqueness, suppose $x = w + y = w' + y'$ for $w, w' \in W, y, y' \in W^\perp$. Then this implies $w - w' = y' - y$, with the left hand side in W and the right hand side in $W^\perp$. But $W \cap W^\perp = \{0_X\}$ by Exercise 3.5(3), so this shows $w - w' = 0_X$, $y' - y = 0_X$, implying $w = w'$ and $y = y'$. ■

Remark 3.8
Proposition 3.7 and its proof give us another view on the orthogonal projection P_W: $P_W(x)$ is the W-component of the unique decomposition of a vector $x \in X$ into $x = w + y$ with $w \in W, y \in W^\perp$.

Proposition 3.9
Let $(X, \langle \cdot, \cdot \rangle)$ be a Hilbert space with $W \subseteq X$ a closed linear subspace. Consider the linear operators $P_W : X \to X$ and $P_{W^\perp} : X \to X$. Then

$$P_W(x) + P_{W^\perp}(x) = x$$

for all $x \in X$, that is, $P_W + P_{W^\perp} = I$, where $I : X \to X$ is the identity map.
Proof:
First, we claim $W \subseteq (W^\perp)^\perp$. Indeed, pick $w \in W$. Then for every $y \in W^\perp$ we have $y \perp w$, so this shows $w \perp W^\perp$, so $W \subseteq (W^\perp)^\perp$. Now, let $x \in X$, and as in Proposition 3.7, write uniquely $x = w + y$ for $w \in W$ and $y \in W^\perp$. But we can rewrite this as $x = y + w$, where $y \in W^\perp$ and $w \in W \subseteq (W^\perp)^\perp$, so by Proposition 3.7, this decomposition into components of the subspaces $W^\perp$ and $(W^\perp)^\perp$ is unique. Therefore, by Remark 3.8 applied twice to the orthogonal decomposition of x with respect to W and with respect to $W^\perp$, we have $w = P_W(x)$ and $y = P_{W^\perp}(x)$, so $x = P_W(x) + P_{W^\perp}(x)$. ■

Corollary 3.10
Let $(X, \langle \cdot, \cdot \rangle)$ be a Hilbert space with $W \subseteq X$ a closed linear subspace. Then $(W^\perp)^\perp = W$.
Proof:
We have $P_W + P_{W^\perp} = I = P_{W^\perp} + P_{(W^\perp)^\perp}$ by Proposition 3.9. Subtracting $P_{W^\perp}$ from both sides, we get $P_W = P_{(W^\perp)^\perp}$. Therefore, $W = \text{im}(P_W) = \text{im}(P_{(W^\perp)^\perp}) = (W^\perp)^\perp$ by Exercise 3.6(3). ■.

Lecture 4: Orthonormal bases for a separable infinite dimensional Hilbert space

Notation and Remark 4.1
(1) Let X be a vector space over $\mathbb{R}$. For every $S \subseteq X$ we use the notation $\text{Span}(S)$ for the linear span of S. This is the set of all possible linear combinations over elements of S, or equivalently the smallest linear subspace of X containing S.
(2) Let $(X, ||\cdot||)$ be a normed vector space over $\mathbb{R}$. For every $S \subseteq X$ we denote $\text{clspan}(S) := \text{cl}(\text{Span}(s))$, where the closure is with respect to the norm $||\cdot||$. Then $\text{clspan}(S)$ is a closed linear subspace of X, where we use the fact that the closure of a linear subspace is still a linear subspace, as can be checked using the sequential description of the closure. Thus, $\text{clspan}(S)$ is the smallest closed linear subspace of X containing S.

Proposition 4.2
Let $(X, ||\cdot||)$ be a normed vector space over $\mathbb{R}$. The following are equivalent:
(1) X is infinite dimensional and separable.
(2) One can find an increasing chain of linear subspaces $X_1 \subseteq \cdots \subseteq X_n \subseteq \ldots$ such that $\dim(X_n) = n$ for all $n \in \mathbb{N}$ and $\bigcup_{n=1}^\infty X_n$ is dense in X.
Proof:
(1) implies (2) is Question 5 of Homework 2, and (2) implies (1) is Question 1 of Homework 3. For the proof of (2) implies (1), it is useful to observe that for the X_n's given in (2), one can find a sequence $(x_n)_{n=1}^\infty$ such that $\{x_1, \ldots, x_n\}$ is a basis for each X_n, $n \in \mathbb{N}$. With this, one can take $Z = \text{Span}_\mathbb{Q}\{x_1, \ldots, x_n, \ldots\}$. ■

Remark 4.3
(1) Consider the sequence $(x_n)_{n=1}^\infty$ discussed in the proof above, with $S = \{x_1, \ldots, x_n, \ldots\} \subseteq X$ (which

has no repetitions by linear independence). Then $\text{clspan}(S) = X$ because $\bigcup_{n=1}^{\infty} X_n$ is dense in X and $\{x_1, \dots, x_n\}$ is a basis for X_n for each $n \in \mathbb{N}$. A sequence with this property is said to be a total sequence in X.
(2) In the setting of Proposition 4.2, all the subspaces X_n are closed linear subspaces of X. This is because, in general, given a normed vector space $(X, ||\cdot||)$ with a finite dimensional linear subspace $V \subseteq X$, V is always closed. This is because V is complete in the metric associated to $||\cdot||$ since it is finite dimensional (which is a consequence of the Extreme Value Theorem), and subsets which are complete in the induced metric are closed.

Quiz
Consider the normed vector space $(C([0,1],\mathbb{R}), ||\cdot||_\infty)$, which is separable.
(a) Give an increasing chain of linear subspaces $X_1 \subseteq X_2 \subseteq \cdots \subseteq X_n \subseteq \dots$ inside $(C([0,1],\mathbb{R}), ||\cdot||_\infty)$ which satisfy (2) of Proposition 4.2.
(b) Give a total sequence $(f_n)_{n=1}^{\infty}$ for the subspaces found in (a).

Proposition 4.4
Let $(X, \langle\cdot,\cdot\rangle)$ be a separable infinite dimensional Hilbert space over $\mathbb{R}$. Let $X_1 \subseteq X_2 \subseteq \cdots \subseteq X_n \subseteq \dots$ be a chain of linear subspaces of X such that $\bigcup_{n=1}^{\infty} X_n$ is dense in X, and $\dim(X_n) = n$ for all $n \in \mathbb{N}$, courtesy of Proposition 4.2. Then we can find a sequence $(\xi_n)_{n=1}^{\infty}$ in X such that
(*) $\text{Span}\{\xi_1, \dots, \xi_n\} = X_n$ for all $n \in \mathbb{N}$.
(**) $\xi_i \perp \xi_j$ for all $i \neq j$.
(***) $||\xi_i|| = 1$ for all $i \in \mathbb{N}$.
A sequence that satisfies (**) and (***) is called an orthonormal sequence in X.
Proof:
Q1(a) of Homework 3 provides us with a sequence $(x_n)_{n=1}^{\infty}$ in X such that, for every $n \in \mathbb{N}$, the vectors $x_1, \dots, x_n$ form a basis for X_n. We will use the Gram-Schmidt orthogonalization procedure to convert the x_n's into an orthonormal sequence. Formally, we proceed by induction on n. For the base case, we put $\xi_1 = \frac{1}{||x_1||} x_1 \in X_1$. Now, inductively suppose that for some $n \geq 1$, we have already constructed orthonormal vectors $\xi_1, \dots, \xi_n$. We look at $X_{n+1} = \text{Span}(x_1, \dots, x_n, x_{n+1})$, and we let

$$\eta := x_{n+1} - \sum_{i=1}^{n} \langle x_{n+1}, \xi_i \rangle \xi_i$$

The Gram-Schmidt process assures us that $\eta \neq 0_X$ with $\langle \eta, \xi_i \rangle = 0$ for all $1 \leq i \leq n$, and $\text{Span}(\xi_1, \dots, \xi_n, \eta) = X_{n+1}$. Now, take $\xi_{n+1} = \frac{\eta}{||\eta||}$ (which is valid since $\eta \neq 0_X$), and the properties obtained above show that $\xi_1, \dots, \xi_n, \xi_{n+1}$ is an orthonormal basis of X_{n+1}, completing the induction. ■

Example 4.5
Consider ℓ^2, as in Homework 1. For the chain of X_n's from Proposition 4.2, we get

$$X_n = \{(x^{(1)}, \dots, x^{(n)}, 0, \dots) : x^{(1)}, \dots, x^{(n)} \in \mathbb{R}\}$$

Then $X_1 \subseteq X_2 \subseteq \cdots \subseteq X_n \subseteq \dots$, with $\dim(X_n) = n$ for all $n \in \mathbb{N}$, and $\bigcup_{n=1}^{\infty} X_n = c_{oo}$, which is dense in ℓ^2. The orthonormal sequence we get is made up of vectors

$$\xi_i = \{0, \dots, 0, \underbrace{1}_{i\text{th pos}}, 0, \dots)$$

for all $i \in \mathbb{N}$. Then observe that $||\xi_i|| = 1$ for all $i \in \mathbb{N}$, $\xi_i \perp \xi_j$ for $i \neq j$, and $\text{Span}(\xi_1, \dots, \xi_n) = X_n$ for all $n \in \mathbb{N}$.

Definition 4.6
Let $(X, \langle\cdot,\cdot\rangle)$ be an infinite dimensional separable Hilbert space. A sequence $(\xi_n)_{n=1}^{\infty}$ as in Proposition 4.4 is called an orthonormal basis (o.n.b.) for $(X, \langle\cdot,\cdot\rangle)$.

Lecture 5: Coefficients with respect to an orthonormal basis

Notation
For this lecture, let $(X, \langle \cdot, \cdot \rangle)$ be a Hilbert space over $\mathbb{R}$ which is infinite dimensional and separable, and fix an orthonormal basis $(\xi_n)_{n=1}^{\infty}$ for it, with corresponding chain of subspaces $X_1 \subseteq X_2 \subseteq \cdots \subseteq X_n \subseteq \ldots$ courtesy of Proposition 4.2.

Definition 5.1
For any $x \in X$, the sequence of real numbers $(\langle x, \xi \rangle)_{n=1}^{\infty}$ is called the "sequence of ξ-coefficients" of x.

Remark 5.2
Let $\mathcal{B} = \{\xi_1, \ldots, \xi_n, \ldots\} \subseteq X$. There are no repetitions in $\mathcal{B}$, and moreover, $||\xi_i - \xi_j|| = \sqrt{2}$ for $i \neq j$ by the Pythagorean theorem. Furthermore, $\text{clspan}(\mathcal{B}) = X$. Indeed, one sees $\bigcup_{n=1}^{\infty} X_n = \text{Span}(\mathcal{B})$, so since $\bigcup_{n=1}^{\infty} X_n$ is dense in X, we get $X = \text{clspan}(\mathcal{B})$. This implies that if $z \in X$ satisfies $z \perp \xi_n$ for all $n \in \mathbb{N}$, we have $z \in 0_X$. Indeed, by Homework 3 Question 2b, this implies $z \in \mathcal{B}^{\perp} = (\text{clspan}(B))^{\perp} = X^{\perp} = \{0_X\}$.

Proposition 5.3
If $x, x' \in X$ have the same sequence of ξ-coefficients, then $x = x'$.
Proof:
For all $n \in \mathbb{N}$, we have $\langle x, \xi_n \rangle = \langle x', \xi_n \rangle$, and so $\langle x - x', \xi_n \rangle = 0$ for all $n \in \mathbb{N}$, which shows by Remark 5.2 that $x - x' = 0_X$. ■

Theorem 5.4
Let $x \in X$, and let $(c_n)_{n=1}^{\infty}$ be the sequence of ξ-coefficients of x. For every $n \in \mathbb{N}$, let $x_n = \sum_{i=1}^{n} c_i \xi_i$. Then,
(1) For all $n \in \mathbb{N}$, $P_{X_n}(x) = x_n$. As a consequence, x_n is the vector in X_n of minimal distance to x.
(2) We have $x_n \to x$ as $n \to \infty$ with respect to $|| \cdot ||$, that is, $\lim_{n\to\infty} ||x_n - x|| = 0$.
Proof:
To prove (1), we first claim that $(x - x_n) \perp \xi_j$ for all $1 \leq j \leq n$. Indeed,

$$\langle x - x_n, \xi_j \rangle = \langle x, \xi_j \rangle - \langle x_n, \xi_j \rangle = c_j - \langle \sum_{i=1}^{n} c_i \xi_i, \xi_j \rangle = c_j - \sum_{i=1}^{n} c_i \langle \xi_i, \xi_j \rangle = c_j - c_j = 0$$

From this, we see that $x - x_n \in \{\xi_1, \ldots, \xi_n\}^{\perp} = (\text{Span}\{\xi_1, \ldots, \xi_n\})^{\perp} = X_n^{\perp}$, using Homework 3 Question 2b. This means that when we write $x = x_n + (x - x_n)$ with $x_n \in X_n, x - x_n \in X_n^{\perp}$, this decomposition is the unique decomposition of x into components in X_n and $X_n^{\perp}$, and so by an equivalent characterization of $P_{X_n}(x)$, we have $x_n = P_{X_n}(x)$, which proves (1).
To prove (2), let $\varepsilon > 0$. Since $\bigcup_{n=1}^{\infty} X_n$ is dense in X, there exists $y \in \bigcup_{n=1}^{\infty} X_n$ such that $||y - x|| < \varepsilon$. By definition of the union, there exists $n_0 \in \mathbb{N}$ such that $y \in X_{n_0}$. Therefore, for any $n \geq n_0$, we have $y \in X_{n_0} \subseteq X_n$, giving that

$$||x - x_n|| = ||x - P_{X_n}(x)|| \leq ||x - y|| < \varepsilon$$

using the definition of $P_{X_n}(x)$ and (1). ■

Remark 5.5
Theorem 5.4(2) says that

$$|| \cdot ||-\lim_{n\to\infty} \left(\sum_{i=1}^{n} c_i \xi_i \right) = x$$

(the notation means the limit with respect to the norm topology of $|| \cdot ||$), which is sometimes written as $x = \sum_{i=1}^{\infty} c_i \xi_i$ and is called a $|| \cdot ||$-convergent series (under an appropriate definition for what is a $|| \cdot ||$-convergent series in X). In this language, Theorem 5.4 is called the "Riesz-Fischer theorem".

Corollary 5.6 (Parseval's formula)
Under the hypotheses of Theorem 5.4, we have

$$||x||^2 = \sum_{i=1}^{\infty} c_i^2$$

Proof:
By Theorem 5.4(2), since $x_n \to x$ as $n \to \infty$ with respect to $|| \cdot ||$, we have $||x_n|| \to ||x||$ (by the reverse triangle inequality). From this, we get

$$||x||^2 = \lim_{n\to\infty} ||x_n||^2 = \lim_{n\to\infty} \left\| \sum_{i=1}^{n} c_i \xi_i \right\|^2 \overbrace{=}^{\text{HW4, Q1}} \lim_{n\to\infty} \sum_{i=1}^{n} c_i^2 = \sum_{i=1}^{\infty} c_i^2 \quad \blacksquare$$

Remark 5.7
Observe that Parseval's formula says that the sequence of ξ-coefficients $c := (c_n)_{n=1}^{\infty}$ is in ℓ^2, since $\sum_{i=1}^{\infty} c_i^2$ is finite, with $||c||_{\ell^2}^2 = ||x||^2$. This gives us a function $\varphi : X \to \ell^2$ that sends $x \in X$ to its sequence of ξ-coefficients, i.e.

$$\varphi(x) = (\langle x, \xi_1 \rangle, \langle x, \xi_2 \rangle, \dots)$$

We will see that φ is a Hilbert space isomorphism from X to ℓ^2. To see this, let us look at some of its properties:
• By bilinearity of the inner product, one sees φ is linear.
• Parseval's formula tells us that φ preserves norms, and so φ is an isometric linear map between Hilbert spaces, as defined on Homework 4, Question 3.
• Our map $\varphi : X \to \ell^2$ has $\varphi(\xi_n) = e_n$ for all $n \in \mathbb{N}$, where

$$e_n = (0, \dots, 0, \underbrace{1}_{n\text{th pos}}, 0, \dots, 0)$$

Indeed, we have

$$\varphi(\xi_n) = (\langle \xi_n, \xi_1 \rangle, \langle \xi_n, \xi_2 \rangle, \dots, \langle \xi_n, \xi_n \rangle, \dots) = (0, 0, \dots, 1, \dots)$$

by orthonormality.
• φ is bijective, hence it is a Hilbert space isomorphism. We get injectivity for free from the fact it is an isometry. To see surjectivity, one must think a bit.
The moral of the story is that X is isomorphic to ℓ^2 as a Hilbert space, written as $X \approx \ell^2$. Hence, any two separable infinite dimensional Hilbert spaces are isomorphic to each other since they are isomorphic to ℓ^2.

Lecture 6: Starting towards the Lebesgue measure on $[a, b]$

Remark 6.1
In Lectures 4 and 5, we had a Hilbert space $(X, \langle \cdot, \cdot \rangle)$ with an orthonormal basis $(\xi_n)_{n=1}^{\infty}$. We want to focus on a special case of X: Start with the space $X_0 = C([-\pi, , \pi], \mathbb{R})$ with inner product $\langle f, g \rangle = \int_{-\pi}^{\pi} f(t)g(t)dt$ and associated norm $||f||_2 = \sqrt{\int_{-\pi}^{\pi} [f(t)]^2 dt}$. This has a nice orthonormal basis, provided by trigonometric functions (see A4Q2). But there's an issue to confront: $(X_0, \langle \cdot, \cdot \rangle)$ is not complete. We will use a Hilbert space $(X, \langle \cdot, \cdot \rangle)$ which is a completion of X_0. But is this X we create a space of functions on $[-\pi, \pi]$? The answer is "almost yes". We will get the space $X = L^2([-\pi, \pi])$, named for the mathematician Lebesgue, where functions $f, g : [-\pi, \pi] \to \mathbb{R}$ are identified when $f(t) = g(t)$ "almost everywhere" on $[-\pi, \pi]$, which means that they agree everywhere except for a "small" set of exceptions, which has a "Lebesgue measure" of 0.

Remark 6.2
What is the Lebesgue measure? It is an advanced version of the notion of length for a subset $S \subseteq [a, b]$.

So, we must identify a convenient collection of subsets $S \subseteq [a,b]$ to which we know how to assign a length $\lambda(S)$.
Example 1: What if S is a subinterval of $[a,b]$, say $S = (c,d]$ with $a \leq c < d \leq b$. It is intuitive that we should assign $\lambda(S) = d - c$.
Example 2: What if S is of the form $S = G \cap [a,b]$, where $G \subseteq \mathbb{R}$ is an open subset of $\mathbb{R}$?
We will use the fact that every open subset G of $\mathbb{R}$ can be uniquely decomposed as $G = \bigcup_{\gamma\in\Gamma} J_\gamma$, where the index set Γ is finite or infinite countable, every J_γ is an open interval, and $J_\gamma \cap J_{\gamma'} = \emptyset$ for $\gamma \neq \gamma'$. Here, the J_γ's are the connected components of G. As a concrete example, take $[a,b] = [0,1]$, with

$$G = (1/2, \infty) \cup (1/8, 1/4) \cup \dots \cup (1/2(4^m), 1/4^m) \cup \dots$$

What length should we assign to $G \cap [0,1]$? Well if we break this up into individual pieces, we get

$$\lambda(S) = \frac{1}{2} + \frac{1}{8} + \frac{1}{32} + \dots + \frac{1}{2\cdot 4^n} + \dots = \frac{1}{2}(1 + \frac{1}{4} + \frac{1}{16} + \dots + \frac{1}{4^n} + \dots) = \frac{1}{2}\frac{1}{1-\frac{1}{4}} = \frac{2}{3}$$

Example 3: Take G as in Example 2, but now look at the compact set $K = [0,1] \setminus G$ (this is compact since it is closed, since we deleted an open set from a closed set, and it is obviously bounded). We could do a calculation along the same spirit as before, since

$$K = [1/4, 1/2] \cup [1/16, 1/8] \cup \dots \cup [1/4^n, 2/4^n] \cup \dots \cup \{0\}$$

But observe that with S as in example 2, we get $K \cap S = \emptyset$ and $K \cup S = [0,1]$. If our notion of length is reasonable, we should get $\lambda(S) + \lambda(K) = \lambda([0,1]) = 1$, so $\lambda(K) = \frac{1}{3}$.

Remark 6.3
With these examples of how length should behave, what are, then, our precise goals for the Lebesgue measure?
Goal 1: Identify a "convenient" collection $\mathfrak{B}$ of subsets of $[a,b]$ that we can work with, in the sense of the above examples. For instance, with the above examples, we should have $S \in \mathfrak{B}$ for every subinterval $S \subseteq [a,b]$, $K \in \mathfrak{B}$ for every compact subset $K \subseteq [a,b]$, and $\mathfrak{B}$ should be compatible with countable unions like the ones we took in the above examples. Such a $\mathfrak{B}$ will be called a sigma-algebra.
Goal 2: Define a function $\lambda : \mathfrak{B} \to [0,\infty)$ such that $\lambda(S)$ serves as the length of $S \in \mathfrak{B}$. This λ will be called a positive measure on $\mathfrak{B}$. Our example above shows that λ should at least satisfy the following property (additive over a countable union): if $S = \bigcup_{n=1}^\infty S_n$ for $S_n \in \mathfrak{B}$ for all $n \in \mathbb{N}$ with $S_n \cap S_m = \emptyset$ for $n \neq m$, then $\lambda(S) = \sum_{n=1}^\infty \lambda(S_n)$.

Lecture 7: Sigma algebras

Notation
For a set Ω, 2^Ω is its powerset (set of all subsets).

Definition 7.1
A collection of sets $\mathfrak{A} \subseteq 2^\Omega$ is a sigma algebra if
(1) $\emptyset \in \mathfrak{A}$
(2) If $A \in \mathfrak{U}$, then $\Omega \setminus A \in \mathfrak{A}$
(3) If $(A_n)_{n=1}^\infty$ is contained in $\mathfrak{U}$, then $\bigcap_{n=1}^\infty A_n \in \mathfrak{A}$.

Remark 7.2
Let $\mathfrak{A} \subseteq 2^\Omega$ be a sigma algebra. These axioms imply that $\mathfrak{A}$ is closed under any operations on countably many sets. For example:
- By (1) and (2), we have $\Omega \in \mathfrak{A}$.
- For finitely many sets $A_1, \dots, A_k \in \mathfrak{A}$, by taking $\Omega = A_n$ for all $n \geq k+1$, we get $\bigcap_{n=1}^k A_n =$

$\bigcap_{n=1}^{\infty} A_n \in \mathfrak{A}$ by (3).
• If $(A_n)_{n=1}^{\infty}$ is contained in $\mathfrak{A}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathfrak{A}$. To see this, we know $\bigcap_{n=1}^{\infty}(\Omega \setminus A_n) \in \mathfrak{A}$, and so by DeMorgan's law, $\bigcup_{n=1}^{\infty} A_n = \Omega \setminus (\bigcap_{n=1}^{\infty}(\Omega \setminus A_n) \in \mathfrak{A}$. • By taking infinitely many copies of the empty set in the property above, if $A_1, \ldots, A_k \in \mathfrak{A}$, then $\bigcup_{n=1}^{k} A_n \in \mathfrak{A}$.
• If $A, B \in \mathfrak{A}$, then $A \setminus B = A \cap (\Omega \setminus B) \in \mathfrak{A}$.

Lemma 7.3
Let $(\mathfrak{A}_i)_{i\in I}$ be a family of sigma algebras of subsets of a nonempty set Ω. Let $\mathfrak{A} = \bigcap_{i\in I} \mathfrak{A}_i$. Then $\mathfrak{A}$ is a sigma algebra.
Proof:
We verify the axioms.
By (1), since $\emptyset \in \mathfrak{A}_i$ for all $i \in I$, it is in their intersection so $\emptyset \in \mathfrak{A}$ so (1) holds.
For (2), pick $A \in \mathfrak{A}$. Then we have $\Omega \setminus A \in \mathfrak{A}_i$ for all $i \in I$ by (2), and so $\Omega \setminus A \in \bigcap_{i\in I} \mathfrak{A}_i = \mathfrak{A}$, so (2) holds.
For (3), pick $(A_n)_{n=1}^{\infty}$ in $\mathfrak{A}$. Then $\bigcap_{n=1}^{\infty} A_n \in \mathfrak{A}_i$ for all $i \in I$. Thus $\bigcap_{n=1}^{\infty} A_n \in \mathfrak{A}$, showing (3). ■

Proposition 7.4
Let Ω be a nonempty set, and let $\mathcal{C} \subseteq 2^{\Omega}$. There exists a unique $\mathfrak{A}_0 \subseteq 2^{\Omega}$ such that
(1) $\mathfrak{A}_0$ is a sigma algebra with $\mathcal{C} \subseteq \mathfrak{A}_0$
(2) Whenever $\mathfrak{A} \subseteq 2^{\Omega}$ is a sigma algebra with $\mathcal{C} \subseteq \mathfrak{A}$, then $\mathfrak{A}_0 \subseteq \mathfrak{A}$.
Proof:
Proving existence, let $(\mathfrak{A}_i)_{i\in I}$ be the family of sigma algebras of subsets of Ω which contain $\mathcal{C}$. Such sigma algebras do exist, since 2^{Ω} is one of them. Let $\mathfrak{A}_0 = \bigcap_{i\in I} \mathfrak{A}_i$. This is a sigma algebra by Lemma 7.3. Moreover, $\mathcal{C} \subseteq \mathfrak{A}_i$ for all $i \in I$, and so $\mathcal{C} \subseteq \bigcap_{i\in I} \mathfrak{A}_i = \mathfrak{A}$, which shows (1) holds. On the other hand, let $\mathfrak{A} \subseteq 2^{\Omega}$ be a sigma algebra such that $\mathcal{C} \subseteq \mathfrak{A}$. Then $\mathfrak{A}$ is counted somewhere among the $\mathfrak{A}_i$'s by definition, so there exists $j \in I$ such that $\mathfrak{A}_j = \mathfrak{A}$. Therefore, we have $\mathfrak{A} \supseteq \bigcap_{i\in I} \mathfrak{A}_i = \mathfrak{A}_0$, which shows (2), proving existence.
For uniqueness, let $\mathfrak{A}_0'$ be a sigma algebra satisfying (1) and (2). But (2) applied to $\mathfrak{A}_0$ (using (1) for $\mathfrak{A}_0'$) implies that $\mathfrak{A}_0 \subseteq \mathfrak{A}_0'$, and with the roles reversed we get $\mathfrak{A}_0' \subseteq \mathfrak{A}_0$, so $\mathfrak{A}_0 = \mathfrak{A}_0'$. ■

Definition 7.5
Let Ω be a nonempty set with $\mathcal{C} \subseteq 2^{\Omega}$. The sigma algebra $\mathfrak{A}_0$ found in Proposition 7.4 is called the sigma algebra generated by $\mathcal{C}$ and will be denoted as σ-Alg($\mathcal{C}$).
An important example of this (as seen on Question 3 of Homework 5) is when (Ω, d) is a metric space and we consider its so-called Borel sigma algebra $\mathcal{B} := \sigma\text{-Alg}(\tau)$, where τ is the set of all open subsets of Ω (its topology).

Lecture 8: Finite positive measures on a sigma algebra

Definition 8.1
Let Ω be a nonempty set, with $\mathfrak{A} \subseteq 2^{\Omega}$ a sigma algebra. A finite positive measure on $\mathfrak{A}$ is a function $\mu : \mathfrak{A} \to [0, \infty)$ such that if we have $A_1, A_2, \ldots, A_n, \cdots \in \mathfrak{A}$ with $A_i \cap A_j = \emptyset$ for $i \neq j$, then $\mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$ (called countable additivity). Notice this is well-defined since $\mathfrak{A}$ is a sigma algebra, so $\bigcup_{n=1}^{\infty} A_n \in \mathfrak{A}$. Moreover, notice this implies $\sum_{n=1}^{\infty} \mu(A_n)$ always converges.

Definition 8.2
Let Ω be a nonempty set with $\mathfrak{A} \subseteq 2^{\Omega}$ a sigma algebra, and μ a finite positive measure on $\mathfrak{A}$. The triple $(\Omega, \mathfrak{A}, \mu)$ is called a finite measure space. If $\mu(\Omega) = 1$, then $(\Omega, \mathfrak{A}, \mu)$ is called a probability space.

Remark
The "finite" is because $\mu(\Omega) < \infty$ by definition of μ. One can consider more general measure spaces if one assigns appropriate meaning to $\mu(\text{set}) = +\infty$.

Remark 8.3 (Properties of finite positive measures)
Let the notation be as above.
(1) $\mu(\emptyset) = 0$. To prove this, let $\alpha = \mu(\emptyset)$. If we take $A_n = \emptyset$ for all $n \in \mathbb{N}$, by countable additivity we have

$$\alpha = \mu(\emptyset) = \mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_N) = \sum_{n=1}^{\infty} \alpha = \lim_{k\to\infty} \sum_{n=1}^{\infty} \alpha = \lim_{k\to\infty} k\alpha$$

This can only converge if $\alpha = 0$.
(2) (Finite additivity) If $A_1, \ldots, A_k \in \mathfrak{A}$ with $A_i \cap A_j = \emptyset$ for $i \neq j$, then $\mu(A_1 \cup \cdots \cup A_k) = \sum_{i=1}^{k} \mu(A_i)$ by countable additivity and using the fact $\mu(\emptyset) = 0$.
(3) (Increasing property) If $A, B \in \mathfrak{A}$ with $A \subseteq B$, then $\mu(A) \leq \mu(B)$. To prove this, let $B = A \cup (B \setminus A)$ with $A, B \setminus A \in \mathfrak{A}$ such that $A \cap (B \setminus A) = \emptyset$, by finite additivity we get

$$\mu(B) = \mu(A) + \underbrace{\mu(B \setminus A)}_{\geq 0} \geq \mu(A)$$

(4) (Complement formula) If $A \in \mathfrak{A}$ with $C = \Omega \setminus A \in \mathfrak{A}$, then $\mu(C) = \mu(\Omega) - \mu(A)$. To prove this, $\Omega = A \cup C$ with $A \cap C = \emptyset$. Then by finite additivity,

$$\mu(\Omega) = \mu(A) + \mu(C)$$

Proposition 8.4 (Continuity along chains)
Let $(\Omega, \mathfrak{A}, \mu)$ be a finite measure space.
(1) Suppose that $(B_n)_{n=1}^{\infty}$ are sets from $\mathfrak{A}$ such that $B_1 \subseteq B_2 \subseteq \cdots \subseteq B_n \subseteq \ldots$, then

$$\mu(\bigcup_{n=1}^{\infty} B_n) = \lim_{n\to\infty} \mu(B_n)$$

(2) Suppose that $(C_n)_{n=1}^{\infty}$ are in $\mathfrak{A}$ such that $C_1 \supseteq C_2 \supseteq \cdots \supseteq C_n \supseteq \ldots$, then

$$\mu(\bigcap_{n=1}^{\infty} C_n) = \lim_{n\to\infty} \mu(C_n)$$

Proof:
Put $A_1 = B_1, A_2 = B_2 \setminus B_1$, $A_3 = B_3 \setminus B_2$, and more generally $A_n = B_n \setminus B_{n-1}$. Then observe that $A_i \in \mathfrak{A}$ for all $i \in \mathbb{N}$, $A_i \cap A_j = \emptyset$ for $i \neq j \in \mathbb{N}$, and $A_1 \cup \cdots \cup A_n = B_n$ for all $n \in \mathbb{N}$. Then this shows that $\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} A_n$, and so

$$\mu(\bigcup_{n=1}^{\infty} B_n) = \mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n) = \lim_{k\to\infty} [\sum_{n=1}^{k} \mu(A_n)] \overbrace{=}^{\text{finite add.}} \lim_{k\to\infty} \mu(\bigcup_{n=1}^{k} A_k) = \lim_{k\to\infty} \mu(B_k)$$

which proves (1).
For (2), we can reduce this to (1) by taking complements. Let $B_m = \Omega \setminus C_m$ for all $n \in \mathbb{N}$. Then this forms a chain $B_1 \subseteq B_2 \subseteq \ldots$, and so by Remark 8.3(4) and (1),

$$\mu(\Omega) - \mu(\bigcap_{n=1}^{\infty} C_n) = \mu(\Omega \setminus \bigcap_{n=1}^{\infty} C_n) = \mu(\bigcup_{n=1}^{\infty} (\Omega \setminus C_n)) = \mu(\bigcup_{n=1}^{\infty} B_n) = \lim_{n\to\infty} \mu(B_n)$$
$$= \lim_{n\to\infty} \mu(\Omega \setminus C_n) = \mu(\Omega) - \lim_{n\to\infty} \mu(C_n)$$

and so

$$\mu(\bigcap_{n=1}^{\infty} C_n) = \lim_{n\to\infty} \mu(C_n) \quad \blacksquare$$

Proposition 8.5
Let $(\Omega, \mathfrak{A}, \mu)$ be a finite measure space. Then,
(1) For any $n \in \mathbb{N}$ and $E_1, \ldots, E_n \in \mathfrak{A}$, we have $\mu(\bigcup_{k=1}^{\infty} E_k) \leq \sum_{k=1}^{n} \mu(E_k)$ (finite sub-additivity).
(2) For any sequence $(E_n)_{n=1}^{\infty}$ contained in $\mathfrak{A}$, we have $\mu(\bigcup_{n=1}^{\infty} E_n) \leq \sum_{n=1}^{\infty} \mu(E_n) \in [0, +\infty]$.
Proof:
For (1), set $A_1 = E_1, A_2 = E_2 \setminus E_1, A_3 = E_3 \setminus (E_1 \cup E_2)$. Then

$$\mu(\bigcup_{k=1}^{n} E_k) = \mu(\bigcup_{k=1}^{n} A_k) = \sum_{k=1}^{n} \mu(A_k) \leq \sum_{k=1}^{n} \mu(E_k)$$

using the fact the A_k's are disjoint and $A_k \subseteq E_k$ for all $k = 1, \ldots, n$. To prove (2), apply the same idea as (1) (exercise). ■

Lecture 9: Regularity

Definition 9.1
Let (Ω, d) be a metric space, and let $\mathcal{T}$ be its topology. Let $\mathfrak{B} = \sigma\text{-Alg}(\mathcal{T})$. $\mathfrak{B}$ is called the Borel sigma algebra of (Ω, d).

Definition 9.2
Let (Ω, d) be a metric space and $\mathfrak{B}$ its Borel sigma algebra. Let $\mu : \mathfrak{B} \to [0, \infty)$ be a finite positive measure. We say μ is regular if for any $S \in \mathfrak{B}$ and $\varepsilon > 0$ then we can find $K, G \in \mathfrak{B}$ where K is compact and G is open, and $K \subseteq S \subseteq G$ with $\mu(G) - \mu(K) < \varepsilon$. We say that μ is closed-regular if for any $S \in \mathfrak{B}$ and $\varepsilon > 0$ we can find $F, G \in \mathfrak{B}$ where F is closed, G is open, and $F \subseteq S \subseteq G$ with $\mu(G) - \mu(F) < \varepsilon$.

Proposition 9.3
Let (Ω, d) be a metric space and $\mathfrak{B}$ its Borel sigma algebra. Let $\mu : \mathfrak{B} \to [0, \infty)$ be a finite positive measure. The following are equivalent:
(1) μ is closed-regular.
(2) For any $S \in \mathfrak{B}$,

$$\mu(S) = \inf\{\mu(G) : G \text{ open with } S \subseteq G\} = \sup\{\mu(F) : F \text{ closed with } F \subseteq S\}$$

Proof:
Proving (1) implies (2), note that for all open sets G such that $S \subseteq G$, we have $\mu(S) \leq \mu(G)$. Therefore, $\mu(S) \leq \inf\{\mu(G) : G \text{ open with } S \subseteq G\}$. To prove the other inequality, for all $n \in \mathbb{N}$ we can take $\varepsilon = 1/n$ and use closed-regularity to find G_n open, F_n closed such that $F_n \subseteq S \subseteq G_n$ with $\mu(G_n) - \mu(F_n) < 1/n$. This tells us that

$$0 \leq \inf\{\mu(G) : G \text{ open with } S \subseteq G\} - \mu(S) \leq \mu(G_n) - \mu(F_n) \leq 1/n$$

so taking the limit as $n \to \infty$, by the squeeze theorem we have $\inf\{\mu(G) : G \text{ open with } S \subseteq G\} = \mu(S)$. A symmetric argument works to show $\mu(S) = \sup\{\mu(F) : F \text{ closed with } F \subseteq S\}$.
Proving (2) implies (1), let $\varepsilon > 0$. Since $\mu(S) = \inf\{\mu(G) : G \text{ open with } S \subseteq G\}$, there exists G open containing S such that $\mu(G) - \mu(S) < \varepsilon/2$, and since $\mu(S) = \sup\{\mu(F) : F \text{ closed with } F \subseteq S\}$ there exists F closed contained in S such that $\mu(S) - \mu(F) < \varepsilon/2$. Combined, this tells us that we have $F \subseteq S \subseteq G$ with

$$\mu(G) - \mu(F) < \varepsilon$$

so μ is closed-regular. ■

Corollary 9.4
Let (Ω, d) be a metric space and $\mathfrak{B}$ its Borel sigma algebra. Let $\mu, \nu : \mathfrak{B} \to [0, \infty)$ be closed-regular

measures on $\mathfrak{B}$ that agree on open sets. Then $\mu = \nu$.
Proof:
Let $S \in \mathfrak{B}$. Then since $\mu(G) = \nu(G)$ when G is open,

$$\mu(S) = \inf\{\mu(G) : G \text{ open with } S \subseteq G\} = \inf\{\nu(G) : G \text{ open with } S \subseteq G\} \quad \blacksquare$$

Theorem 9.5
Let (Ω, d) be a metric space, $\mathfrak{B}$ its Borel sigma algebra, and $\mu : \mathfrak{B} \to [0, \infty)$ any finite positive measure. Then μ is closed regular.
Proof:
We use the following method, which as an Andu-ism is called "the method of friendly sets". Let

$$\mathcal{F} = \{S \in \mathfrak{B} : \forall \varepsilon > 0, \text{there exist } F \text{ closed, } G \text{ open with } F \subseteq S \subseteq G \text{ and } \mu(G) - \mu(F) < \varepsilon\}$$

We have $\mathcal{F} \subseteq \mathfrak{B}$ by definition, our goal is to prove that $\mathfrak{B} \subseteq \mathcal{F}$ so that $\mathfrak{B} = \mathcal{F}$. We will first prove that every open set is in $\mathcal{F}$ (so $\mathcal{T} \subseteq \mathcal{F}$, where $\mathcal{T}$ is (Ω, d)'s topology). Then, we will prove $\mathcal{F}$ is a sigma-algebra. Therefore, since $\mathcal{F}$ is a sigma-algebra containing $\mathcal{T}$, by Proposition 7.4 we will get $\sigma\text{-Alg}(\mathcal{T}) \subseteq \mathcal{F}$, but $\sigma\text{-Alg}(\mathcal{T}) = \mathfrak{B}$, completing the proof.
To prove the first claim, let $G \subseteq \Omega$ be open, and let $\varepsilon > 0$. We must find $F', G' \subseteq \Omega$ such that F' is closed, G' is open, with $F' \subseteq G \subseteq G'$ with $\mu(G') - \mu(F') < \varepsilon$. Indeed, recall from PMath 351, that G as an open subset of a metric space, is a set of type F_σ, so there exists an increasing chain of closed sets $F_1 \subseteq F_2 \subseteq \dots$ such that $\bigcup_{n=1}^\infty F_n = G$. By continuity of μ along chains, we get $\lim_{n\to\infty} \mu(F_n) = \mu(G)$. Therefore, we can find $n_0 \in \mathbb{N}$ such that $\mu(F_{n_0}) > \mu(G) - \varepsilon$. Let $F' = F_{n_0}, G' = G$. Hence $F' \subseteq G \subseteq G'$, and

$$\mu(G') - \mu(F') = \mu(G) - \mu(F_{n_0}) < \varepsilon$$

which proves the claim. To prove the second claim ($\mathcal{F}$ is a sigma-algebra), we verify the definition. Clearly $\emptyset \in \mathcal{F}$ since it is clopen (so take $F = G = \emptyset$). To prove it is closed under complements, let $B \in \mathcal{F}$, and consider $\Omega \setminus B$. Since $B \in \mathcal{F}$, there exist F' closed and G' open so that $F' \subseteq B \subseteq G'$ satisfying $\mu(G') - \mu(F') < \varepsilon$. Let $G = \Omega \setminus F'$, which is open, and $F = \Omega \setminus G'$ which is closed. Then we get $F \subseteq \Omega \setminus B \subseteq G$, and

$$\mu(G) - \mu(F) = \mu(\Omega) - \mu(F') - (\mu(\Omega) - \mu(G')) = \mu(G') - \mu(F') < \varepsilon$$

so $\Omega \setminus B \in \mathcal{F}$, so $\mathcal{F}$ is closed under complements. To show it is closed under countable intersection, we first claim $\mathcal{F}$ is closed under finite intersections. It suffices to prove it is closed under intersection by two sets, the claim then follows by induction. To prove this, let $B_1, B_2 \in \mathcal{F}$, and let $\varepsilon > 0$. Since $B_1 \in \mathcal{F}$, there exists F_1 closed and G_1 open such that $F_1 \subseteq B_1 \subseteq G_1$ with $\mu(G_1) - \mu(F_1) < \varepsilon/2$. Similarly for $B_2 \in \mathcal{F}$, there exist F_2 closed and G_2 open such that $F_2 \subseteq B_2 \subseteq G_2$ with $\mu(G_2) - \mu(F_2) < \varepsilon/2$. Then, for $G := G_1 \cap G_2$ and $F := F_1 \cap F_2$ we have

$$F \subseteq B_1 \cap B_2 \subseteq G$$

with G open and F closed. Moreover, we have

$$\mu(G) - \mu(F) = \mu(G \setminus F) = \mu((G_1 \cap G_2) \setminus (F_1 \cap F_2)) \overbrace{\leq}^{(G_1\cap G_2)\setminus(F_1\cap F_2)\subseteq(G_1\setminus F_1)\cup(G_2\setminus F_2)} \mu((G_1 \setminus F_1) \cup (G_2 \setminus F_2))$$
$$\leq \mu(G_1 \setminus F_1) + \mu(G_2 \setminus F_2) < \varepsilon/2 + \varepsilon/2 = \varepsilon$$

which proves the claim.
We now claim that $\mathcal{F}$ is closed under countable intersections of decreasing chains $C_1 \supseteq C_2 \supseteq \cdots \supseteq \dots$. Let $C := \bigcap_{n=1}^\infty C_n$, and let $\varepsilon > 0$. For each $n \in \mathbb{N}$, by definition of $\mathcal{F}$ there exists F_n closed and G_n open such that $F_n \subseteq C_n \subseteq G_n$ with $\mu(G_n \setminus F_n) < \dfrac{\varepsilon}{2 \cdot 2^n}$. Then, we have

$$\bigcap_{n=1}^\infty F_n \subseteq C \subseteq \bigcap_{n=1}^\infty G_n$$

Then,

$$\mu(\bigcap_{n=1}^{\infty} G_n \setminus \bigcap_{n=1}^{\infty} F_n) \leq \mu(\bigcup_{n=1}^{\infty} G_n \setminus F_n) < \varepsilon/2$$

The only issue is that $\bigcap_{n=1}^{\infty} G_n$ is of type G_δ and not necessarily open. To fix this issue, recall we can write any G_δ set as the intersection $\bigcap_{n=1}^{\infty} G_n = \bigcap_{n=1}^{\infty} D_n$ with $D_1 \supseteq D_2 \supseteq \dots$ all open, and so $\bigcap_{n=1}^{\infty} G_n \subseteq D_k$ for all $k \in \mathbb{N}$. Then by continuity along chains, $\mu(\bigcap_{n=1}^{\infty} G_n) = \mu(\bigcap_{n=1}^{\infty} D_n) = \lim_{k\to\infty} \mu(D_k)$, so

$$\mu(D_k \setminus \bigcap_{n=1}^{\infty} G_n) = \mu(D_k) - \mu(\bigcap_{n=1}^{\infty} G_n)$$

so taking the limit $k \to \infty$, since the rightmost term does not depend on k, we have

$$0 = \mu(\bigcap_{n=1}^{\infty} D_n) - \mu(\bigcap_{n=1}^{\infty} G_n) = \lim_{k\to\infty} \mu(D_k) - \mu(\bigcap_{n=1}^{\infty} G_n) = \lim_{k\to\infty} (\mu(D_k) - \mu(\bigcap_{n=1}^{\infty} G_n))$$

so there exists $k_0 \in \mathbb{N}$ such that $\mu(D_{k_0}) - \mu(\bigcap_{n=1}^{\infty} G_n) < \varepsilon/2$, with D_{k_0} open. Thus we have

$$\bigcap_{n=1}^{\infty} F_n \subseteq C \subseteq \bigcap_{n=1}^{\infty} G_n \subseteq D_{k_0}$$

with

$$\mu(D_{k_0}) - \mu(\bigcap_{n=1}^{\infty} F_n) < \varepsilon/2 + \mu(\bigcap_{n=1}^{\infty} G_n) - \mu(\bigcap_{n=1}^{\infty} F_n) < \varepsilon/2 + \varepsilon/2 = \varepsilon$$

which proves the claim.
From the preceding claims, we prove that $\mathcal{F}$ is closed under all countable intersections: for all $n \in \mathbb{N}$, define $C_n = B_1 \cap \dots \cap B_n$. Then $C_n \in \mathcal{F}$ by closure under finite intersections, and we have $C_1 \supseteq C_2 \supseteq \dots$ with $\bigcap_{n=1}^{\infty} C_n = \bigcap_{n=1}^{\infty}(B_1 \cap \dots \cap B_n) = \bigcap_{n=1}^{\infty} B_n$, so $\bigcap_{n=1}^{\infty} B_n = \bigcap_{n=1}^{\infty} C_n \in \mathcal{F}$ by the previous claim, which finishes the proof that $\mathcal{F}$ is a sigma-algebra and thus the proof. ■

Corollary 9.6
Let (Ω, d) be a metric space and $\mathfrak{B}$ its Borel sigma algebra. Let $\mu, \nu : \mathfrak{B} \to [0, \infty)$ be finite positive measures on $\mathfrak{B}$ that agree on open sets. Then $\mu = \nu$.
Proof:
Corollary 9.4 combined with Theorem 9.5. ■

Lecture 10: The Lebesgue measure on an interval $[a, b]$

Definition 10.1
Let $a < b \in \mathbb{R}$, and consider $\Omega = [a, b]$ as a metric space under the usual (induced) Euclidean distance, with $\mathfrak{B}$ its Borel sigma algebra. We say a finite positive measure $\mu : \mathfrak{B} \to [0, \infty)$ is length-fitting when it measures the open balls of Ω according to their length: $\mu((\alpha, \beta)) = \beta - \alpha$ for all $a \leq \alpha \leq \beta \leq b$, $\mu([a, \beta)) = \beta - a$ for all $a < \beta \leq b$, $\mu((\alpha, b]) = b - \alpha$ for all $a \leq \alpha < b$, and $\mu([a, b]) = b - a$.

Theorem/Definition 10.2
Given $\Omega = [a, b]$ and $\mathfrak{B}$ its Borel sigma algebra, there exists a unique finite positive measure $\mu : \mathfrak{B} \to [0, \infty)$ that is length-fitting, which we call the Lebesgue measure on $[a, b]$.

Remark 10.3
Uniqueness follows from A6Q4, which is a consequence of closed-regularity and the structure theorem for open subsets of a closed interval. We will now prove existence.

Notation 10.4
Let $\mathcal{U}$ be the set of open balls in $[a,b]$. Define $\lambda_{ball} : U \to [0,\infty)$ by $\lambda_{ball}(U) = \sup(U) - \inf(U)$ for $U \in \mathcal{U}$. Note that $0 \leq \lambda_{ball}(U) \leq b-a$.

Notation 10.5
Let $\mathcal{T}$ be the topology of $\Omega = [a,b]$ with respect to the usual distance d. Define $\lambda_{open} : \mathcal{T} \to [0,\infty)$ as follows: first, $\lambda_{open}(\emptyset) = 0$. Then, for $G \neq \emptyset \in \mathcal{T}$, put

$$\lambda_{open}(G) = \sup\{\sum_{i=1}^{n} \lambda_{ball}(U_i) : n \in \mathbb{N}, U_1, \dots, U_n \in \mathcal{U} \ s.t. \ U_1, \dots, U_n \subseteq G, U_i \cap U_j = \emptyset \ \ \forall i \neq j\}$$

Remark 10.6
Suppose $U_1, \dots, U_n \in \mathcal{U}$ with $U_i \cap U_j = \emptyset$ for all $i \neq j$. Then we can reorder the U_i's so that

$$\sup(U_1) \leq \inf(U_2) \leq \sup(U_2) \leq \inf(U_3) \leq \cdots \leq \sup(U_{n-1}) \leq \inf(U_n)$$

which is called the canonical ordering of $U_1, \dots, U_n$.

Remark 10.7
(1) For $G \neq \emptyset \in \mathcal{T}$, we have $\lambda_{open}(G) > 0$. To see this, let $x \in G$ and let $r > 0$ such that $B(x;r) \subseteq G$. Then

$$\lambda_{open}(G) \geq \sum_{i=1}^{n} \lambda_{ball}(B(x;r)) \geq \sup(B(x;r)) - \inf(B(x;r)) > 0$$

(2) For $G_1 \subseteq G_2 \in \mathcal{T}$, we have $\lambda_{open}(G_1) \leq \lambda_{open}(G_2)$: this is clear if $G_1 = \emptyset$ (which occurs exactly when $\lambda_{open}(G_1) = 0$), otherwise the supremum defining $\lambda_{open}(G_2)$ uses a finer collection of numbers than the supremum defining $\lambda_{open}(G_1)$, so $\lambda_{open}(G_2) \geq \lambda_{open}(G_1)$.
(3) For $U \in \mathcal{U} \subseteq \mathcal{T}$, we have $\lambda_{open}(U) = \lambda_{ball}(U)$. To see this, in the definition of λ_{open} we can take $n = 1$ and $U_1 = U$ to see

$$\lambda_{open}(U) \geq \sum_{i=1}^{n} \lambda_{ball}(U_i) = \lambda_{ball}(U_1) = \lambda_{ball}(U)$$

Conversely, it suffices (by definition of λ_{open} as a supremum) to prove that if we have $U_1, \dots, U_n \in \mathcal{U}$ such that $U_1, \dots, U_n \subseteq U$ with $U_i \cap U_j = \emptyset$ for $i \neq j$, then $\sum_{i=1}^{n} \lambda_{ball}(U_i) \leq \lambda_{open}(U)$. Relabel $U_1, \dots, U_n$ into canonical ordering by Remark 10.6, and use canonical ordering to get

$$\sum_{i=1}^{n} \lambda_{ball}(U_i) = \sum_{i=1}^{n} [\sup(U_i) - \inf(U_i)] = (\sup(U_n) - \inf(U_n)) + (\sup(U_{n-1}) - \inf(U_{n-1}) + \cdots + (\sup(U_1) - \inf(U_1))$$

$$\leq (\sup(U_n) - \inf(U_n)) + (\inf(U_n) - \sup(U_{n-1})) + (\sup(U_{n-1}) - \inf(U_{n-1})) + \cdots + (\sup(U_1) - \inf(U_1))$$
$$= \sup(U_n) - \inf(U_1) \leq \sup(U) - \inf(U) = \lambda_{open}(U)$$

which proves the claim (we rearranged the series via canonical ordering to make it telescope).
(4) $\lambda_{open}(G) \leq b-a$ for all $G \in \mathcal{T}$. Indeed, by (2), we get

$$\lambda_{open}(G) \leq \lambda_{open}([a,b]) = \lambda_{ball}([a,b]) = b-a$$

Fact
Let $G, G' \in \mathcal{T}$ such that $G \cap G' = \emptyset$. Suppose $U \in \mathcal{U}$ is such that $U \subseteq G \cup G'$. Then either $U \subseteq G$ or $U \subseteq G'$. To see this, we will prove either $U \cap G = \emptyset$ or $U \cap G' = \emptyset$. The proof goes by contradiction (and just follows by connectedness of U): assume for a contradiction that $U \cap G \neq \emptyset \neq U \cap G'$. Then $U \cap G'$ and $U \cap G$ are open sets in the metric space (U, d_0) where d_0 is the induced Euclidean distance (this is just the definition of the subspace topology). Then we have $(U \cap G) \cap (U \cap G') = U \cap (G \cap G') = \emptyset$ since

$G \cap G' = \emptyset$, and $(U \cap G) \cup (U \cap G') = U \cap (G \cup G') = U$ since $G \cup G'$ contains U, so this shows (U, d_0) is disconnected, which is impossible since U is an interval.

Proposition 10.8
Let $G_1, G_2 \in \mathcal{T}$ be such that $G_1 \cap G_2 = \emptyset$. Then $\lambda_{open}(G_1 \cup G_2) = \lambda_{open}(G_1) + \lambda_{open}(G_2)$.
Proof:
We may assume without loss of generality that $G_1, G_2 \neq \emptyset$. We will prove $\lambda_{open}(G_1 \cup G_2) \leq \lambda_{open}(G_1) + \lambda_{open}(G_2)$ and $\lambda_{open}(G_1 \cup G_2) \geq \lambda_{open}(G_1) + \lambda_{open}(G_2)$. To see the first inequality, by definition of $\lambda_{open}(G_1 \cup G_2)$ as a supremum, it suffices to check that if we have $U_1, \dots, U_K \subseteq G_1 \cup G_2$ satisfying $U_i \cap U_j = \emptyset$ for $i \neq j$, then

$$\sum_{i=1}^{n} \lambda_{ball}(U_i) \leq \lambda_{open}(G_1) + \lambda_{open}(G_2)$$

To see this, by the Fact above, for each $i \in \{1, \dots, k\}$, we have either $U_i \in G_1$ or $U_i \in G_2$. Thus, we may split $\{1, \dots, k\}$ disjointly into subsets I_1, I_2 with $U_i \in G_1$ for all $i \in I_1$ and $U_j \in G_2$ for all $j \in I_2$, allowing us to write

$$\sum_{i=1}^{k} \lambda_{ball}(U_i) = \sum_{i \in I_1} \lambda_{ball}(U_i) + \sum_{i \in I_2} \lambda_{ball}(U_i) \leq \lambda_{open}(G_1) + \lambda_{open}(G_2)$$

where the last inequality follows by definition of the supremum we are taking λ_{open} over (with respect to G_1 and G_2 for the respective sums over I_1 and I_2), which proves this direction.
To prove the other direction, it suffices to show that for all $\varepsilon > 0$,

$$\lambda_{open}(G_1 \cup G_2) > \lambda_{open}(G_1) + \lambda_{open}(G_2) - \varepsilon$$

Thus, let $\varepsilon > 0$. By definition of the supremum, there exist $U_1, \dots, U_n \in \mathcal{U}$ contained in G_1 with trivial intersection satisfying

$$\sum_{i=1}^{n} \lambda_{ball}(U_i) > \lambda_{open}(G_1) - \varepsilon/2$$

and $U_1', \dots, U_m'$ contained in G_2 with trivial intersection satisfying

$$\sum_{i=1}^{m} \lambda_{ball}(U_i) > \lambda_{open}(G_2) - \varepsilon/2$$

Then, by definition of the supremum in $\lambda_{open}(G_1 \cup G_2)$, we get

$$\lambda_{open}(G_1 \cup G_2) \geq \sum_{i=1}^{n} \lambda_{ball}(U_i) + \sum_{i=1}^{m} \lambda_{ball}(U_i') > \lambda_{open}(G_1) + \lambda_{open}(G_2) - \varepsilon$$

which proves the claim. ■

Example
Let $\Omega = [0, 1]$, and let G be the complement of the ternary Cantor set C. This is the countable union of disjoint open intervals (the middle thirds deleted to form the Cantor set). We claim $\lambda_{open}(G)$ is the countable sum of the λ_{ball}'s of the open intervals. Indeed, by definition of the supremum we know $\lambda_{open}(G)$ is greater than or equal to any finite sum of the lengths of these open intervals. It turns out the lengths of these intervals are a geometric sequence convering to 1, so $\lambda_{open}(G) \geq 1$, and $\lambda_{open}(G) \leq 1$ by Remark 10.7, so $\lambda_{open}(G) = 1$.

Lecture 11: The Lebesgue measure on a compact interval II

Notation and Remark 11.1
Following the notation of Lecture 10 (so $\Omega = [a, b]$ is a compact interval with d the Euclidean distance), let $\mathcal{K}$ be the set of compact subsets of Ω, equivalently the set of closed sets of Ω. Define $\lambda_{cpct} : \mathcal{K} \to [0, \infty)$ by $\lambda_{cpct}(K) = (b-a) - \lambda_{open}(\Omega \setminus K)$ for all $K \in \mathcal{K}$.

Remark 11.2
Because Ω is connected, $\mathcal{T} \cap \mathcal{K} = \{\emptyset, \Omega\}$, and we know $\lambda_{open}(\emptyset) = 0, \lambda_{open}(\Omega) = b - a$, which implies $\lambda_{cpct}(\emptyset) = 0$ and $\lambda_{cpct}(\Omega) = b - a$.

Definition 11.3
We will say $G \in \mathcal{T}$ is elementary when it can be written as a finite union $G = U_1 \cup \cdots \cup U_m$ of disjoint open balls $U_1, \ldots, U_m \in \mathcal{U}$.

Remark 11.4
Let $G = U_1 \cup \cdots \cup U_m$ be an elementary open set. Then by Proposition 10.8, $\lambda_{open}(G) = \sum_{i=1}^m \lambda_{ball}(U_i)$.

Lemma 11.5
Suppose $K \in \mathcal{K}$ and $G \in \mathcal{T}$ with $K \subseteq G$. Then we can write $G = G' \cup G''$ for $G', G'' \in \mathcal{T}$ such that $G' \cap G'' = \emptyset$ with G' elementary and $K \subseteq G'$.
Proof:
If G is elementary, take $G' = G$ and $G'' = \emptyset$ and we are done. Otherwise, suppose G is not elementary, so we may write G as a countable disjoint union of its connected components $G = \bigcup_{n=1}^\infty U_n$ for $U_n \in \mathcal{U}$ for all $n \in \mathbb{N}$. Then $K \subseteq \bigcup_{n=1}^\infty U_n$, so by compactness, relabelling if necessary, there exist $U_1, \ldots, U_k$ such that $K \subseteq \bigcup_{n=1}^k U_n$. Take $G' = \bigcup_{n=1}^k U_n$, which is indeed elementary and contains K, and take $G'' = \bigcup_{n=k+1}^\infty U_n$. These choices suffice because our original decomposition $G = \bigcup_{n=1}^\infty U_n$ is a disjoint union. ■

Lemma 11.6
Let $K \in \mathcal{K}$ and $G \in \mathcal{T}$ be such that G is elementary and $K \subseteq G$. Let $D = G \setminus K \in \mathcal{T}$. Then $\lambda_{open}(D) + \lambda_{cpct}(K) = \lambda_{open}(G)$.
Proof:
On A7b. ■

Proposition 11.7
Let $K \in \mathcal{K}$ and $G \in \mathcal{T}$ be such that $K \subseteq G$. Let $D = G \setminus K \in \mathcal{T}$. Then $\lambda_{open}(D) + \lambda_{cpct}(K) = \lambda_{open}(G)$.
Proof:
By Lemma 11.5, write $G = G' \cup G''$ for G', G'' disjoint with $K \subseteq G'$, where $\lambda_{open}(G) = \lambda_{open}(G') + \lambda_{open}(G'')$ by Proposition 10.8, and $D = (G' \setminus K) \cup G''$, so $\lambda_{open}(D) = \lambda_{open}(G' \setminus K) + \lambda_{open}(G'')$. Now by Lemma 11.6 applied to K within G',

$$\lambda_{open}(D) + \lambda_{cpct}(K) = \lambda_{open}(G'') + \lambda_{open}(G' \setminus K) + \lambda_{cpct}(K) = \lambda_{open}(G'') + \lambda_{open}(G') = \lambda_{open}(G) \quad ■$$

Proposition 11.7 tells us we can calculate lengths of compact sets K by calculating the difference $\lambda_{open}(G) - \lambda_{open}(D)$ for any open set G containing K, which generalizes Notation 11.1.

Corollary 11.8
If $K \in \mathcal{K}$ and $G \in \mathcal{T}$, then $\lambda_{cpct}(K) \leq \lambda_{open}(G)$.
Proof:
In the context of Proposition 11.7, we have $\lambda_{open}(D) \geq 0$. ■

Definition 11.9
Let $\mathfrak{B} = \sigma\text{-Alg}(\mathcal{T})$ be the Borel sigma algebra of Ω. We say $B \in \mathfrak{B}$ is friendly (this is non-standard) if

for every $\varepsilon > 0$, there exists compact K and open G such that $K \subseteq B \subseteq G$ with $\lambda_{open}(G) - \lambda_{cpct}(K) < \varepsilon$. We denote the set of friendly sets by $\mathfrak{F}$.

Remark 11.10
Suppose we are able to show $\mathfrak{F}$ is a sigma-algebra that contains all open balls (which is indeed true). Then it follows that $\mathfrak{B} = \mathfrak{F}$.

Remark 11.11
Here is another approach to define the length of sets we have seen that is equivalent (for a reference, see Rudin Real and Complex Analysis, Chapter 2). We use the same "friendly" machinery but define λ_{open} and λ_{cpct} differently. We set $\widetilde{\lambda}_{open}(G)$ to be the supremum of all Riemann integrals over $[a, b]$ of continuous functions $f : [a, b] \to [0, 1]$ supported in G, and set $\widetilde{\lambda}_{cpct}(K)$ to be the infimum of all continuous functions $f : [a, b] \to [0, 1]$ with $f(K) = \{1\}$.

Lecture 12: Measurable Spaces, Measurable Functions, and the space $\mathrm{Bor}(\Omega, \mathbb{R})$

Definition 12.1
A measurable space is a pair (Ω, σ) where Ω is a nonempty set and σ is a sigma algebra of subsets of Ω. We are interested in the case where Ω is a metric space with $\sigma = \mathfrak{B}$.

Definition 12.2
Let (Ω, σ) and (Γ, α) be measurable spaces. We say a function $f : \Omega \to \Gamma$ is σ/α measurable if $f^{-1}(A) \in \sigma$ for all $A \in \alpha$.

Lemma 12.3
Let $(\Omega, \sigma), (\Gamma, \alpha), (\Lambda, \beta)$ be measurable spaces. Then if $f : \Omega \to \Gamma$ is σ/α measurable and $g : \Gamma \to \Lambda$ is α/β measurable, then $g \circ f$ is σ/β measurable.
Proof:
Let $B \in \beta$. Then $g^{-1}(B) \in \alpha$ by measurability, and $(g \circ f)^{-1}(B) = f^{-1}(g^{-1}(B)) \in \sigma$ by measurability again. ■

Lemma 12.4 ("Give Me a Break Lemma")
Let (Ω, σ) and (Γ, α) be measurable spaces where α is generated by some $\beta \subseteq \alpha$. Let $f : \Omega \to \Gamma$ be a function and suppose that $f^{-1}(B) \in \sigma$ for all $B \in \beta$. Then f is σ/α measurable.
Proof:
Let $\mathcal{F} = \{A \in \alpha : f^{-1}(A) \in \sigma\}$. We want to show $\mathcal{F} = \alpha$. Note that $\beta \subseteq \mathcal{F}$. We will show $\mathcal{F}$ is a sigma algebra. We have $\emptyset \in \mathcal{F}$ since $f^{-1}(\emptyset) = \emptyset \in \sigma$. Next, let $A \in \mathcal{F}$. Then $\Omega \setminus f^{-1}(A) = f^{-1}(\Gamma \setminus A)$, but $\Omega \setminus f^{-1}(A) \in \sigma$ since σ is a sigma algebra with $f^{-1}(A) \in \sigma$, which shows $\mathcal{F}$ is closed under complements. To show $\mathcal{F}$ is closed under countable intersections, let $(A_n)_{n=1}^{\infty}$ be a sequence of sets in $\mathcal{F}$. Then $f^{-1}(\bigcap_{n=1}^{\infty} A_n) = \bigcap_{n=1}^{\infty} f^{-1}(A_n)$, so this is a countable intersection of elements of σ and thus is in σ, which shows $\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}$.
Altogether, we have $\beta \subseteq \mathcal{F}$ with $\mathcal{F}$ a sigma-algebra, so because α is generated by β, this shows $\alpha \subseteq \mathcal{F}$. Since $\mathcal{F} \subseteq \alpha$ for free, we have $\mathcal{F} = \alpha$ which completes the proof. ■

Proposition 12.5
Let $(\Omega, d), (\Gamma, d')$ be metric spaces. Let $\mathfrak{B}_\Omega$ and $\mathfrak{B}_\Gamma$ be their Borel sigma algebras. Let $f : \Omega \to \Gamma$ be continuous. Then f is $\mathfrak{B}_\Omega/\mathfrak{B}_\Gamma$ measurable (such a function is called a Borel function).
Proof:
Let $\mathcal{T}_\Omega$ and $\mathcal{T}_\Gamma$ be the topologies of (Ω, d) and (Γ, d') respectively. For a continuous function f, by con-

tinuity we have $f^{-1}(G') \in \mathcal{T}_\Omega \subseteq \mathfrak{B}_\Omega$ for all $G' \in \mathcal{T}_\Gamma$. Therefore, by Lemma 12.4, since $\mathfrak{B}_\Gamma$ is generated by $\mathcal{T}_\Gamma$, this shows f is $\mathfrak{B}_\Omega/\mathfrak{B}_\Gamma$ measurable. ■

Notation 12.6
Let (Ω, σ) be a measurable space. We denote $\mathrm{Bor}(\Omega, \mathbb{R}) := \{f : \Omega \to \mathbb{R} : f \text{ is } \sigma/\mathfrak{B}_\mathbb{R} \text{ measurable}\}$ to be the set of Borel functions from Ω to $\mathbb{R}$.

Corollary 12.7
Proposition 12.5 says that, for a metric space (Ω, d) considered with its Borel sigma algebra, we have $\mathcal{C}(\Omega, \mathbb{R}) \subseteq \mathrm{Bor}(\Omega, \mathbb{R})$. ■

Lemma 12.8
Let (Ω, σ) be a measurable space. Let $k \in \mathbb{N}$, and consider a function $f : \Omega \to \mathbb{R}^k$, with $f_1, \ldots, f_k : \Omega \to \mathbb{R}$ its component functions. Then f is $\sigma/\mathfrak{B}_{\mathbb{R}^k}$ measurable if and only if each f_j is $\sigma/\mathfrak{B}_\mathbb{R}$ measurable for $j = 1, \ldots, k$.
Proof:
In the forward direction, fix $j \in \mathbb{N}$. Let $\pi_j : \mathbb{R}^k \to \mathbb{R}$ be the jth projection map, which is continuous and thus $\mathfrak{B}_{\mathbb{R}^k}/\mathfrak{B}_\mathbb{R}$ measurable. Then $f_j = \pi_j \circ f$, so by Lemma 12.3 we get that f_j is $\sigma/\mathfrak{B}_\mathbb{R}$ measurable, proving this direction.
Let $\mathfrak{C}$ be the collection of open cubes in $\mathbb{R}^k$. On the homework, we prove that $\sigma\text{-Alg}(\mathfrak{C}) = \mathfrak{B}_{\mathbb{R}^k}$. Therefore, by Lemma 12.4, it suffices to prove that $f^{-1}(C) \in \sigma$ for all $C \in \mathfrak{C}$. Pick a cube $C = (a_1 - r, a_1 + r) \times \cdots \times (a_k - r, a_k + r) \in \mathfrak{C}$. Compute

$$f^{-1}(C) = \{x \in \Omega : f(x) \in C\} = \{x \in \Omega : f_i(x) \in (a_i - r, a_i + r), 1 \le i \le k\} = \bigcap_{i=1}^{k} f_i^{-1}((a_i - r, a_i + r))$$

Since each component is $\sigma/\mathfrak{B}_{\mathbb{R}^k}$ measurable and we are taking a finite intersection (which σ is closed under), we have $f^{-1}(C) \in \sigma$. ■

Proposition 12.9 Let (Ω, σ) be a measurable space. Let $f, g : \Omega \to \mathbb{R} \in \mathrm{Bor}(\Omega, \mathbb{R})$. Then $f + g, fg, \min(f, g)$ and $\max(f, g)$ are in $\mathrm{Bor}(\Omega, \mathbb{R})$ (so it is an algebra of functions).
Proof:
We prove the case for the product, the rest are similar. Let $h : \mathbb{R}^2 \to \mathbb{R}$ be given by $h(x, y) = xy$. Then $(fg)(x) = h(f(x), g(x))$, so if we let $F : \Omega \to \mathbb{R}$ be given by $F(x) = (f(x), g(x))$ for $x \in \Omega$, we have $fg = h \circ F$. h is continuous and thus $\mathfrak{B}_{\mathbb{R}^2}/\mathfrak{B}_\mathbb{R}$ measurable, and the component functions of F are measurable so it is by the previous proposition, so the composition $fg = h \circ F$ is $\sigma/\mathfrak{B}_\mathbb{R}$ measurable. ■

Remark 12.10
Note that if $f \in \mathrm{Bor}(\Omega, \mathbb{R})$, then $cf, |f| \in \mathrm{Bor}(\Omega, \mathbb{R})$ for $c \in \mathbb{R}$ by continuity of scalar multiplication and absolute value.

Lecture 13: Convergence in $\mathrm{Bor}(\Omega, \mathbb{R})$

Proposition 13.1
Let $(f_n)_{n=1}^\infty$ be a sequence of functions in $\mathrm{Bor}(\Omega, \mathbb{R})$ that converge pointwise to $f : \Omega \to \mathbb{R}$. Then $f \in \mathrm{Bor}(\Omega, \mathbb{R})$.

This follows from the stronger statement:

Proposition 13.2
Let $(f_n)_{n=1}^\infty$ be a sequence contained in $\mathrm{Bor}(\Omega, \mathbb{R})$ such that for all $x \in \Omega$, we have $f(x) = \limsup_{n\to\infty} f_n(x)$

for some $f : \Omega \to \mathbb{R}$. Then $f \in \mathrm{Bor}(\Omega, \mathbb{R})$. The analogous statement for lim inf's holds by taking negatives.
Sketch of Proof:
We prove the lim sup version. We have $f(x) = \inf_{k\geq 1} \sup_{n\geq k} f_n(x)$ for all $x \in \Omega$. Write $h_k(x) = \sup_{n\geq k} f_n(x)$ for each $k \in \mathbb{N}$, so $f(x) = \inf_{k\geq 1} h_k(x)$ for all $x \in \Omega$ (as a minor detail, we need to check that the h_k's give values in $\mathbb{R}$ and are not infinite). Lemma 13.4(1) gives that $h_k \in \mathrm{Bor}(\Omega, \mathbb{R})$ for each $k \in \mathbb{N}$, so by Lemma 13.4(2) we get $f \in \mathrm{Bor}(\Omega, \mathbb{R})$. ■

These are proved with the following lemma (note that $\limsup_{n\to\infty} f_n(x) = \lim_{n\to\infty} \sup_{k\geq n} f_k(x) = \inf_{n\geq 1} \sup_{k\geq n} f_k(x)$).

Lemma 13.4
(1) Let $f : \Omega \to \mathbb{R}$ and suppose $(f_m)_{m=1}^{\infty}$ is contained in $\mathrm{Bor}(\Omega, \mathbb{R})$ such that $f(x) = \sup\{f_m(x) : m \in \mathbb{N}\}$. Then $f \in \mathrm{Bor}(\Omega, \mathbb{R})$.
(2) Let $g : \Omega \to \mathbb{R}$ and suppose $(g_m)_{m=1}^{\infty}$ is contained in $\mathrm{Bor}(\Omega, \mathbb{R})$ such that $g(x) = \inf\{g_m(x) : m \in \mathbb{N}\}$. Then $g \in \mathrm{Bor}(\Omega, \mathbb{R})$.
Proof:
We start with (1). By an assignment question, it suffices to prove that for all $t \in \mathbb{R}$, $\{x \in \Omega : f(x) \leq t\} \in \sigma$. Equivalently, $f^{-1}((-\infty, t]) \in \sigma$ (since sets of this form generate the Borel sigma algebra). Fix $t \in \mathbb{R}$. Then,

$$x \in f^{-1}((-\infty, t]) \iff f(x) \leq t \iff \sup_{m\in\mathbb{N}} f_m(x) \leq t \iff f_m(x) \leq t, \forall m \in \mathbb{N} \iff x \in \bigcap_{m=1}^{\infty} f_m^{-1}((-\infty, t])$$

Since each f_m is Borel we have $f_m^{-1}((-\infty, t]) \in \sigma$ for all $m \in \mathbb{N}$, so $f^{-1}((-\infty, t]) \in \sigma$, which proves (1). To prove (2), take negatives of (1) (using the fact $\mathrm{Bor}(\Omega, \mathbb{R})$ is an algebra of functions). ■

Lecture 14: Integration on $\mathrm{Bor}^+(\Omega, \mathbb{R})$

Notation and Remark 14.1
For this lecture, fix a measurable space (Ω, σ, μ).
Let $\mathrm{Bor}^+(\Omega, \mathbb{R}) := \{f \in \mathrm{Bor}(\Omega, \mathbb{R}) : f(x) \geq 0 \forall x \in \Omega\}$. Note that $\mathrm{Bor}^+(\Omega, \mathbb{R})$ is closed under taking minimums (denoted by a wedge). It has essentially the same properties as $\mathrm{Bor}(\Omega, \mathbb{R})$, except it is only closed under non-negative scalar multiplication (so it isn't a vector space). We write $f \leq g$ to mean $f(x) \leq g(x)$ for all $x \in \Omega$.

Definition 14.2
We say a measurable scale (not a standard term) is a system $\tau = \begin{pmatrix} A_1 & \dots & A_m \\ \alpha_1 & \dots & \alpha_m \end{pmatrix}$ where $m \in \mathbb{N}$, $\alpha_i \in [0, \infty), A_i \in \sigma$ with the A_i's pairwise disjoint for $1 \leq i \leq m$. For such a τ we define its weight to be $W(\tau) := \sum_{i=1}^{m} \alpha_m \mu(A_m)$.

Definition 14.3
Let $f \in \mathrm{Bor}^+(\Omega, \mathbb{R})$ and let $\tau = \begin{pmatrix} A_1 & \dots & A_m \\ \alpha_1 & \dots & \alpha_m \end{pmatrix}$ be a measurable scale. We write $\tau \prec f$ if for each $i = 1, \dots, m$, for all $x \in A_i$ we have $\alpha_i \leq f(x)$.

Definition 14.4
Let $f \in \mathrm{Bor}^+(\Omega, \mathbb{R})$ and let τ be some measurable scale such that $\tau \prec f$. Such a τ always exists by taking $\alpha_i = 0$ for all $i = 1, \dots, m$. It thus makes sense to define $L^+(f) = \sup\{W(\tau) : \tau \prec f\}$, possibly allowing the case $L^+(f) = +\infty$.

Remark 14.5
$L^+ : \mathrm{Bor}^+(\Omega,\mathbb{R}) \to [0,\infty]$ is one way to define the Lebesgue integral, where it is usually denoted $L^+(f) =: \int_\Omega f(x)d\mu(x)$ (which we will switch to eventually).

Remark 14.6
If $f \leq g \in \mathrm{Bor}^+(\Omega,\mathbb{R})$, then for any measurable scale τ such that $\tau \prec f$ we have $\tau \prec g$, so taking supremums we get $L^+(f) \leq L^+(g)$.

Remark 14.7
If we have $f \in \mathrm{Bor}^+(\Omega,\mathbb{R})$ and $\alpha \in (0,\infty)$, then $\alpha f \in \mathrm{Bor}^+(\Omega,\mathbb{R})$, and $\alpha L^+(f) = L^+(\alpha f)$ (with the convention $\alpha\cdot\infty = \infty$). We obtain this by scaling the α_i's used in the definition of a measurable scale $\tau = \begin{pmatrix} A_1 & \dots & A_m \\ \alpha_1 & \dots & \alpha_m \end{pmatrix} \prec f$ by α to get a new measurable scale $\tau^* = \begin{pmatrix} A_1 & \dots & A_m \\ \alpha\alpha_1 & \dots & \alpha\alpha_m \end{pmatrix} S \prec \alpha f$. We can extend this and take $\alpha = 0$, since $\tau \prec \alpha f = 0$ forces $W(\tau) = 0$, which gives $0 = L^+(\alpha f)$. By convention, we take $0\cdot\infty = 0$ so $\alpha L^+(f) = L^+(\alpha f)$ for all non-negative scalars α.

Proposition 14.8
Let $f, g \in \mathrm{Bor}^+(\Omega,\mathbb{R})$. Then $L^+(f+g) \geq L^+(f) + L^+(g)$.
Proof:
If $L^+(f+g) = \infty$, then the result is clear, so assume $L^+(f+g) < \infty$. This implies that $L^+(f), L^+(g) < \infty$ since $f \leq f+g$ and $g \leq f+g$ (since they are non-negative functions).
Now, let $\varepsilon > 0$. We will show $L^+(f+g) > L^+(f) + L^+(g) - \varepsilon$, from which the result follows by letting $\varepsilon \to 0$. By definition, we know $L^+(f) = \sup\{W(\tau) : \tau \prec f\}$. Thus, let $\tau_1 = \begin{pmatrix} A_1 & \dots & A_k \\ \alpha_1 & \dots & \alpha_k \end{pmatrix} \prec f$ be a measurable scale such that $W(\tau_1) > L^+(f) - \varepsilon/2$, and $\tau_2 = \begin{pmatrix} B_1 & \dots & B_\ell \\ \beta_1 & \dots & \beta_\ell \end{pmatrix} \prec g$ be a measurable scale such $W(\tau_2) > L^+(g) - \varepsilon/2$. Without loss of generality, we may assume $A_1 \cup \dots \cup A_k = \Omega$ and $B_1 \cup \dots \cup B_\ell = \Omega$. Indeed, if not, we may replace τ_1 by $\widetilde{\tau}_1 = \begin{pmatrix} A_1 & \dots & A_k & \Omega \setminus \bigcup_{j=1}^k A_j \\ \alpha_1 & \dots & \alpha_k & 0 \end{pmatrix}$ (and similarly for τ_2). Now, consider the measurable scale

$$\rho := \begin{pmatrix} A_1\cap B_1 & A_1\cap B_2 & \dots & A_1\cap B_\ell & \dots & A_k\cap B_1 & \dots & A_k\cap B_\ell \\ \alpha_1+\beta_1 & \alpha_1+\beta_2 & \dots & \alpha_1+\beta_\ell & \dots & \alpha_k+\beta_1 & \dots & \alpha_k+\beta_\ell \end{pmatrix}$$

ρ is indeed a measurable scale because for $(i,j) \neq (i',j')$ we have $(A_i\cap B_j)\cap(A_{i'}\cap B_{j'}) = (A_i\cap A_{i'})\cap(B_j\cap B_{j'}) = \emptyset$ since either $i \neq i'$ implies $A_i \cap A_{i'} = \emptyset$ or $j \neq j'$ implies $B_j \cap B_{j'} = \emptyset$.
We claim $\rho \prec f+g$. Indeed, let $(i,j) \in \{1,\dots,k\}\times\{1,\dots,\ell\}$ and let $x \in A_i \cap B_j$. Then $x \in A_i$ implies (since $\tau_1 \prec f$) that $f(x) \geq \alpha_i$, and $x \in G_j$ implies (since $\tau_2 \prec g$) that $g(x) \geq \beta_j$. Add these to get $(f+g)(x) = f(x)+g(x) \geq \alpha_i + \beta_j$, which shows $\rho \prec f+g$.
Next, we claim $W(\rho) = W(\tau_1) + W(\tau_2)$. Indeed,

$$W(\rho) = \sum_{1\leq i\leq k, 1\leq j\leq\ell} (\alpha_i+\beta_j)\mu(A_i\cap B_j) = \overbrace{\sum_{1\leq i\leq k, 1\leq j\leq\ell} \alpha_i\mu(A_i\cap B_j)}^{=:S_1} + \overbrace{\sum_{1\leq i\leq k, 1\leq j\leq\ell} \beta_j\mu(A_i\cap B_j)}^{=:S_2}$$

We compute

$$S_1 = \sum_{1\leq i\leq k, 1\leq j\leq\ell} \alpha_i\mu(A_i\cap B_j) = \sum_{i=1}^k\sum_{j=1}^\ell \alpha_i\mu(A_i\cap B_j) = \sum_{i=1}^k \alpha_i \sum_{j=1}^\ell \mu(A_i\cap B_j)$$

Since for a fixed $i \in \{1,\dots,k\}$, the sets $A_i\cap B_1,\dots,A_i\cap B_\ell$ are pairwise disjoint with

$$\bigcup_{j=1}^\ell (A_i\cap B_j) = A_i \cap (\bigcup_{j=1}^\ell B_j) = A_i \cap \Omega = A_i$$

and so inserting this into the above sum combined with the additivity of μ gives

$$S_1 = \sum_{i=1}^{k} \alpha_i \sum_{j=1}^{\ell} \mu(A_i \cap B_j) = \sum_{i=1}^{k} \alpha_i \mu(A_i) = W(\tau_1)$$

and similarly $S_2 = W(\tau_2)$. From our original sum, this gives $W(\rho) = W(\tau_1) + W(\tau_2)$, proving the claim. With this, we conclude that

$$L^+(f+g) \overbrace{\geq}^{\rho \prec f+g} W(\rho) = W(\tau_1) + W(\tau_2) > L^+(f) + L^+(g) - \varepsilon \quad \blacksquare$$

Remark 14.9
We would like to show the stronger statement $L^+(f+g) = L^+(f) + L^+(g)$. But the proof of Proposition 14.8 does not work in this case. Typically to show this, a different approach is used: one approximates L^+ with simple functions and shows it for those functions.

Lecture 15: Integration in $\mathrm{Bor}^+(\Omega, \mathbb{R})$ II: Approximation with Simple Functions

Definition 15.1
A function $f : \Omega \to \mathbb{R}$ is simple if $f(\Omega)$ is finite. Denote $\mathrm{Bor}_S^+(\Omega, \mathbb{R})$ to be the set of simple functions in $\mathrm{Bor}^+(\Omega, \mathbb{R})$.

Remark 15.2
Let $f \in \mathrm{Bor}_S^+(\Omega, \mathbb{R})$. Then there are nonempty disjoint $M_1, \dots, M_k \in \sigma$ and $\alpha_1 > \cdots > \alpha_k > 0$ such that $f = \sum_{i=1}^{k} \alpha_i \chi_{M_i}$.

Fact 15.3
(1) Let $f \in \mathrm{Bor}_S^+(\Omega, \mathbb{R})$ and write $f = \sum_{i=1}^{k} \alpha_i \chi_{M_i}$. Then $L^+(f) = \sum_{i=1}^{k} \alpha_i \mu(M_i)$. Moreover, if $f = \sum_{j=1}^{\ell} \beta_j \chi_{N_j}$ for $N_1, \dots, N_\ell$ disjoint measurable sets, then $L^+(f) = \sum_{j=1}^{\ell} \beta_j \mu(N_j)$.
(2) If $f, g \in \mathrm{Bor}_S^+(\Omega, \mathbb{R})$, then $L^+(f+g) = L^+(f) + L^+(g)$.
(3) Let $f \in \mathrm{Bor}(\Omega, \mathbb{R})$. Then there is a sequence $(f_n)_{n=1}^\infty$ contained in $\mathrm{Bor}_S^+(\Omega)$ with $f_1 \leq f_2 \leq \cdots \leq f_n \leq \dots$ and $\lim_{n\to\infty} f_n(x) = f(x)$ for all $x \in \Omega$.
(4) Let $f \in \mathrm{Bor}(\Omega, \mathbb{R})$ and let $(f_n)_{n=1}^\infty$ be as in (3). Then $L^+(f_1) \leq L^+(f_2) \leq \cdots \leq L^+(f_n) \leq \dots$ and $\lim_{n\to\infty} L^+(f_n) = L^+(f)$.
Proof of (2):
Write $f = \sum_{i=1}^{k} \alpha_i \chi_{M_i}$ and $g = \sum_{j=1}^{\ell} \beta_j \chi_{N_j}$. We will do induction on ℓ, so $g = \beta \chi_N$ for some $N \in \sigma$, $\beta > 0$. Let $M = \bigcup_{i=1}^{k} M_i$ and put $N_i = M_i \cap N$ for $1 \leq i \leq k$, and $D_i = M_i \setminus N$. Moreover, put $N_0 = N \setminus M$ and $D = \Omega \setminus (M \cup N)$. Then Ω is the disjoint union

$$\Omega = \bigcup_{i=1}^{k} N_i \cup \bigcup_{i=1}^{k} D_i \cup N_0 \cup D$$

Let $h = f + g = \sum_{i=1}^{k} \alpha_i \chi_{M_i} + \beta \chi_N$. Then working casewise, we have $h(x) = \alpha_i + \beta$ if $x \in N_i$ for $i > 0$, $h(x) = \alpha_i$ if $x \in D_i$ for $i > 0$, $h(x) = \beta$ if $x \in N_0$, and $h(x) = 0$ if $x \in D$. Therefore,

$$h = \sum_{i=1}^{k} (\alpha_i + \beta) \chi_{N_i} + \sum_{i=1}^{k} \alpha_i \chi_{D_i} + \beta \chi_{N_0}$$

so using (1),

$$L^+(f+g) = L^+(h) = \sum_{i=1}^{k} (\alpha_i + \beta) \mu(N_i) + \sum_{i=1}^{k} \alpha_i \mu(D_i) + \beta \mu(N_0)$$

$$= \sum_{i=1}^{k} \alpha_i \underbrace{(\mu(N_i) + \mu(D_i))}_{=\mu(M_i)} + \beta \underbrace{\sum_{i=0}^{k} \mu(N_i)}_{=\mu(N)} = \sum_{i=1}^{k} \alpha_i \mu(M_i) + \beta\mu(N) = L^+(f) + L^+(g)$$

The inductive step is similar. ■

Proposition 15.4
Let $f, g \in \mathrm{Bor}^+(\Omega, \mathbb{R})$. Then $L^+(f+g) = L^+(f) + L^+(g)$.
Proof:
Use (3) above to get sequences $(f_n)_{n=1}^{\infty}, (g_n)_{n=1}^{\infty}$ of functions in $\mathrm{Bor}_S^+(\Omega, \mathbb{R})$ such that $f_1 \leq f_2 \leq \dots$ and $g_1 \leq g_2 \leq \dots$ converging pointwise to f and g respectively. Let $h_n = f_n + g_n$ for each $n \in \mathbb{N}$. Then $h_n \in \mathrm{Bor}_S^+(\Omega, \mathbb{R})$ for each $n \in \mathbb{N}$ and converges pointwise to $f+g$. Then by (4) above, $L^+(f_n) \to L^+(f)$, $L^+(g_n) \to L^+(g)$, and $L^+(h_n) \to L^+(h)$. By fact 2, $L^+(h_n) = L^+(f_n) + L^+(g_n)$, so taking limits we get $L^+(f+g) = L^+(f) + L^+(g)$. ■

Lecture 16: The space $\mathcal{L}^1(\mu)$

Fix a finite measure space (Ω, σ, μ). Recall we have a map $L^+ : \mathrm{Bor}^+(\Omega, \mathbb{R}) \to [0, \infty]$. Our goal is to find a second incarnation of this function to work with functions in $\mathrm{Bor}(\Omega, \mathbb{R})$ (without the positivity assumption). This turns out not to work so well, so we will choose a suitable subspace $\mathcal{L}^1(\mu) \subseteq \mathrm{Bor}(\Omega, \mathbb{R})$.

Definition 16.1
For $f \in \mathrm{Bor}(\Omega, \mathbb{R})$ we define its positive and negative parts to be $f_+, f_- : \Omega \to \mathbb{R}$ by $f_+(x) = \max\{f(x), 0\}$, $f_-(x) = \max\{-f(x), 0\}$ for $x \in \Omega$.

Remark 16.2
Let $f \in \mathrm{Bor}(\Omega, \mathbb{R})$.
(1) $f_+ \in \mathrm{Bor}^+(\Omega, \mathbb{R})$: it is Borel by Proposition 12.9, and non-negative by definition.
(2) Similarly, $f_- \in \mathrm{Bor}^+(\Omega, \mathbb{R})$.
(3) We have $f_+ + f_- = |f|$, and $f_+ - f_- = f$.

Proposition and Definition 16.3
For $f \in \mathrm{Bor}(\Omega, \mathbb{R})$, we have $L^+(|f|) < \infty$ if and only if $L^+(f_+) < \infty$ and $L^+(f_-) < \infty$. If f satisfies this condition, then f is called integrable with respect to μ, and we define its (Lebesgue) integral to be $\int_\Omega f d\mu := L^+(f^+) - L^+(f_-) \in \mathbb{R}$.
Proof:
In the forward direction, by Remark 14.6, we have $L^+(f_+) \leq L^+(|f|) < \infty$ and $L^+(f_-) \leq L^+(|f|) < \infty$, which proves this direction. Conversely, by Proposition 15.4 and Remark 16.2, we have $L^+(|f|) = L^+(f_+) + L^+(f_-)$, so if $L^+(f_+), L^+(f_-) < \infty$ then the right hand side is a finite sum. ■

Notation 16.4
We define $\mathcal{L}^1(\mu)$ to be the set of all integrable functions in $\mathrm{Bor}(\Omega, \mathbb{R})$.

Proposition 16.5
$\mathcal{L}^1(\mu)$ is a linear subspace of $\mathrm{Bor}(\Omega, \mathbb{R})$.
Proof:
First, it is easy to check the zero function is in $\mathcal{L}^1(\mu)$.
We first claim $\mathcal{L}^1(\mu)$ is closed under addition. Let $f, g \in \mathcal{L}^1(\mu)$. We have

$$|f+g| \leq |f| + |g| \implies L^+(|f+g|) \leq L^+(|f| + |g|) = L^+(|f|) + L^+(|g|) < \infty$$

where the last inequality is because $L^+(|f|), L^+(|g|) < \infty$, which proves the claim.
To prove $\mathcal{L}^1(\mu)$ is closed under scalar multiplication, let $f \in \mathcal{L}^1(\mu)$ and $\alpha \in \mathbb{R}$. Then $L^+(|\alpha f|) = |\alpha| L^+(|f|) < \infty$. ■

Our next goal is to prove the map $I : \mathcal{L}^1(\mu) \to \mathbb{R}$ given by $f \mapsto \int_\Omega f d\mu$ is linear.

Lemma 16.6
Let $f \in \mathcal{L}^1(\mu)$ and write $f = h_1 - h_2$ for $h_1, h_2 \in \mathrm{Bor}^+(\Omega, \mathbb{R})$ such that $L^+(h_1), L^+(h_2) < \infty$ (an example of which is $h_1 = f_+, h_2 = f_-$). Then $\int_\Omega f d\mu = L^+(h_1) - L^+(h_2)$.
Proof:
Write $f = f_+ - f_- = h_1 - h_2$, so that $f_+ + h_2 = f_- + h_1$, where both sides are in $\mathrm{Bor}^+(\Omega, \mathbb{R})$. Taking L^+, we get $L^+(f_+ + h_2) = L^+(f_- + h_1)$, so $L^+(f_+) + L^+(h_2) = L^+(f_-) + L^+(h_1)$. Because everything is finite by Proposition 16.3, rearranged we get $\int_\Omega f d\mu = L^+(f_+) - L^+(f_-) = L^+(h_1) - L^+(h_2)$. ■

Proposition 16.7
(1) Let $f, g \in \mathcal{L}^1(\mu)$. Then $\int_\Omega (f+g) d\mu = \int_\Omega f d\mu + \int_\Omega g d\mu$.
(2) Let $f \in \mathcal{L}^1(\mu), \alpha \in \mathbb{R}$. Then $\int_\Omega \alpha f d\mu = \alpha \int_\Omega f d\mu$.
Proof:
(1) Write $f = f_+ - f_-, g = g_+ - g_-$, so $f + g = (f_+ + g_+) - (f_- + g_-)$. Then by the lemma,

$$\int_\Omega (f+g) d\mu = L^+(f_+ + g_+) - L^+(f_- + g_-) = L^+(f_+) + L^+(g_+) - L^+(f_-) - L^+(g_-)$$

$$= L^+(f_+) - L^+(f_-) + L^+(g_+) - L^+(g_-) = \int_\Omega f d\mu + \int_\Omega g d\mu$$

where everything is justified since everything is finite by Proposition 16.3, which proves (1).
(2) From the definition, we have $\alpha f = \alpha f_+ - \alpha f_-$. Thus if $\alpha > 0$, we get $\int_\Omega \alpha f d\mu = L^+(\alpha f_+) - L^+(\alpha f_-) = \alpha(L^+(f_+) - L^+(f_-)) = \alpha \int_\Omega f d\mu$. If $\alpha = 0$, we get that everything is zero is immediately, and if $\alpha = -1$, then $(-f)_+ = f_-$ and $(-f)_- = f_+$ and the rest follows. Since we can write any real number as zero, positive, or -1 times a positive number, we are done. ■

Remark 16.8
For $f \in \mathrm{Bor}^+(\Omega, \mathbb{R})$ we have $f \in \mathcal{L}^1(\mu)$ if and only if $L^+(f) < \infty$ and if this does hold, then $\int_\Omega f d\mu = L^+(f) \in [0, \infty)$. As such, it is common convention to write $\int_\Omega f d\mu$ instead of $L^+(f)$, even when $L^+(f) = +\infty$. Therefore, we have two notions of the Lebesgue integral: $\int_\Omega f d\mu = L^+(f) \in [0, \infty]$ for $f \in \mathrm{Bor}^+(\Omega, \mathbb{R})$ and $\int_\Omega f d\mu \in \mathbb{R}$ for $f \in \mathcal{L}^1(\mu)$. As seen above, these two notions agree on their overlap.

Remark 16.9
(1) For $f, g \in \mathcal{L}^1(\mu)$, if $f \leq g$ then $\int_\Omega f d\mu \leq \int_\Omega g d\mu$.
(2) For $f \in \mathcal{L}^1(\mu)$, we have $|\int_\Omega f d\mu| \leq \int_\Omega |f| d\mu < \infty$.
To prove (1), by linearity of the integral, we have $\int_\Omega g d\mu - \int_\Omega f d\mu = \int_\Omega (g - f) d\mu = L^+(g - f) \in [0, \infty)$, so $\int_\Omega g d\mu - \int_\Omega f d\mu \geq 0$.
To prove (2), use the fact that $-|f| \leq f \leq |f|$, so by (1), $-L^+(|f|) = \int_\Omega -|f| d\mu \leq \int_\Omega f d\mu \leq \int_\Omega |f| d\mu = L^+(|f|)$, hence $\int_\Omega f d\mu \in [-L^+(f), L^+(|f|)]$, so $|\int_\Omega f d\mu| \leq L^+(|f|)$.

Lecture 17: The space $\mathcal{L}^2(\mu)$ and its variation $L^2(\mu)$

Notation 17.1
We denote $\mathcal{L}^2(\mu) = \{f \in \mathrm{Bor}(\Omega, \mathbb{R}) : \int_\Omega f^2 d\mu < \infty\} = \{f \in \mathrm{Bor}(\Omega, \mathbb{R}) : f^2 \in \mathcal{L}^1(\mu)\}$.

Proposition 17.2
$\mathcal{L}^2(\mu)$ is a linear subspace of $\mathrm{Bor}(\Omega, \mathbb{R})$.

Proof:
It is clear that $0 \in \mathcal{L}^2(\mu)$. To verify $\mathcal{L}^2(\mu)$ is closed under vector addition, let $f+g \in \mathcal{L}^2(\mu)$. Then $(f+g)^2 \leq 2(f^2+g^2)$ (since $(f-g)^2 \geq 0$, then expand), so $\int_\Omega (f+g)^2 d\mu \leq \int_\Omega 2(f^2+g^2)d\mu = 2\int_\Omega f^2 d\mu + 2\int_\Omega g^2 d\mu < \infty$, so $f+g \in \mathcal{L}^2(\mu)$. To verify $\mathcal{L}^2(\mu)$ is closed under scalar multiplication, let $f \in \mathcal{L}^2(\mu)$ and $\alpha \in \mathbb{R}$. Then

$$\int_\Omega (\alpha f)^2 d\mu = \int_\Omega \alpha^2 f^2 d\mu = \alpha^2 \int_\Omega f^2 d\mu < \infty \quad \blacksquare$$

Proposition and Definition 17.3
If $f, g \in \mathcal{L}^2(\mu)$, then $fg \in \mathcal{L}^1(\mu)$. It thus makes sense to define $\langle f, g\rangle := \int_\Omega fg d\mu \in \mathbb{R}$.
Proof:
We have $|fg| \leq \frac{1}{2}(f^2+g^2)$ (from $(|f|-|g|)^2 \geq 0$), hence

$$\int_\Omega |fg| d\mu \leq \frac{1}{2}\int_\Omega f^2 d\mu + \frac{1}{2}\int_\Omega g^2 d\mu < \infty$$

so $fg \in \mathcal{L}^1(\mu)$. $\blacksquare$

Remark 17.4
A consequence of the above proposition is that $\mathcal{L}^2(\mu) \subseteq \mathcal{L}^1(\mu)$. This is because we can take $g = 1$, which is indeed in $\mathcal{L}^1(\mu)$: $\int_\Omega 1^2 d\mu = \int_\Omega 1 d\mu = \mu(\Omega) < \infty$, where the last equality is because 1 is a simple function, and the finiteness is because μ is a finite measure.

Proposition 17.5
$\langle \cdot, \cdot \rangle$ defined above on $\mathcal{L}^2(\mu)$ has the following properties:
(1) It is bilinear.
(2) It is symmetric.
(3) It is non-negative definite. That is, $\langle f, f\rangle \geq 0$ for all $f \in \mathcal{L}^2(\mu)$.
Proof:
Exercise. $\blacksquare$

Remark
We do not get positive-definiteness in (3), and indeed this does not hold: $\langle f, f\rangle = 0$ does not imply $f = 0$ (take an indicator function on a set of measure zero). Thus, $(\mathcal{L}^2(\mu), \langle \cdot, \cdot\rangle)$ is not an inner product space. Nevertheless, some properties still hold:

Notation 17.6
For $f \in \mathcal{L}^2(\mu)$, we denote $||f||_2 := \sqrt{\langle f, f\rangle} \in [0, \infty)$.

Proposition 17.7
(1) Cauchy-Schwarz holds for $\langle \cdot, \cdot\rangle$, that is, $|\langle f, g\rangle| \leq ||f||_2 ||g||_2$.
(2) The map $\mathcal{L}^2(\mu) \to \mathbb{R}$, $f \mapsto ||f||_2$ is a seminorm.
Proof:
(1) The typical proof of Cauchy-Schwarz does not depend on positive-definiteness of the inner product, so it works as normal.
(2) The usual proofs for a norm induced by an inner product apply mutatis mutandis to the case where our "inner product" is non-negative definite. $\blacksquare$

Let's figure out how to deal with this seminorm. Suppose we have a vector space $\mathcal{V}$ over $\mathbb{R}$ equipped with a seminorm $||\cdot||$.

Lemma 17.8
Let $\mathcal{N} = \{v \in \mathcal{V} : ||v|| = 0\}$ be the "nullspace" of $||\cdot||$. Then $\mathcal{N}$ is a linear subspace of $\mathcal{V}$.

Proof:
It contains the zero vector because $||0|| = |0| \cdot ||v|| = 0$ for all $v \in \mathcal{V}$.
It is closed under vector addition by the triangle inequality: let $v, w \in \mathcal{N}$. Then $0 \leq ||v + w|| \leq ||v|| + ||w|| = 0 + 0 = 0$, so $||v + w|| = 0$.
It is closed under scalar multiplication by homogeneity: for $\alpha \in \mathbb{R}, v \in \mathcal{N}$, $||\alpha v|| = |\alpha| ||v|| = |\alpha| 0 = 0$. ■

Notation 17.9
Let $\mathcal{N} \subseteq \mathcal{V}$ be as in Lemma 17.8, and let $\mathcal{Q} := \mathcal{V}/\mathcal{N}$ be the quotient vector space, and let $\pi : \mathcal{V} \to \mathcal{Q}$ be the quotient map. For any $v \in \mathcal{V}$, we will denote $\widehat{v} := \pi(v) \in \mathcal{Q}$.

Proposition and Definition 17.10
Let $\mathcal{N} \subseteq \mathcal{V}$ and $\mathcal{Q}$ be as above. Then $|| \cdot ||$ descends to $\mathcal{Q}$, on which it is a norm.
Proof:
The fact that $|| \cdot ||$ descends in a well-defined manner is because $\mathcal{N}$ is its nullspace, using the standard argument from algebra. It forms a seminorm on the quotient because of the properties upstairs, and is a full norm because we quotiented out by the nullspace. ■

Now, we return to our context where (Ω, σ, μ) is a finite measure space, and consider the vector space $\mathcal{L}^2(\mu)$ with its seminorm $|| \cdot ||_2$.

Notation 17.11
Let $\mathcal{N} = \{f \in \mathcal{L}^2(\mu) : ||f||_2 = 0\}$. We denote $L^2(\mu) = \mathcal{L}^2(\mu)/\mathcal{N} = \{\widehat{f} : f \in \mathcal{L}^2(\mu)\}$ where $\widehat{f} = \widehat{g}$ if and only if $f - g \in \mathcal{N}$. Then Proposition 17.10 shows $|| \cdot ||_2$ descends to a norm, still denoted $|| \cdot ||_2$, on $L^2(\mu)$.

Definition 17.12
Let $f, g \in \mathrm{Bor}(\Omega, \mathbb{R})$. We say that f and g are equal almost everywhere (with respect to μ), denoted "equal a.e.-μ", to meant that there exists $S \in \sigma$ with $\mu(S) = 0$ such that $f(x) = g(x)$ for all $x \in \Omega \setminus S$.

Proposition 17.13
Let $\mathcal{N}$ be the nullspace of the seminorm $|| \cdot ||_2$ on $\mathcal{L}^2(\mu)$. Then $\mathcal{N} = \{f \in \mathrm{Bor}(\Omega, \mathbb{R}) : f = 0 \text{ a.e.-}\mu\}$.

Remark 17.14
By Proposition 17.13, for $f, g \in \mathcal{L}^2(\mu)$, we now see that $\widehat{f} = \widehat{g}$ if and only if $f = g$ a.e.-μ.

Lecture 18: Back to Hilbert spaces: the case of $L^2[-\pi, \pi]$

We look at the finite measure space (Ω, σ, μ), where $\Omega = [-\pi, \pi]$, $\sigma = \mathfrak{B}_{[-\pi,\pi]}$ is its Borel sigma algebra, and $\mu = \mu_{[-\pi,\pi]}$ is the Lebesgue measure. Consider the associated space $L^2(\mu)$, which in this context is usually denoted $L^2[-\pi, \pi]$. This gives us a normed vector space $(L^2[-\pi, \pi], || \cdot ||_2)$. How does this compare with the Banach space $(C([-\pi, \pi], \mathbb{R}), || \cdot ||_\infty)$?

Remark 18.1
If $f \in C[-\pi, \pi]$, then $f \in \mathcal{L}^2[-\pi, \pi]$, with $||f||_2 \leq \sqrt{2\pi} ||f||_\infty$. First, note that $f \in \mathrm{Bor}(\Omega, \mathbb{R})$ by continuity, and denote $\alpha = ||f||_\infty$. Then $f^2 \leq \alpha^2 1$, so integrating both sides, we get

$$||f||_2^2 = \int_{[-\pi,\pi]} f^2 d\mu \leq \int_{[-\pi,\pi]} \alpha^2 d\mu = \alpha^2 \mu([-\pi, \pi]) = \alpha^2 \cdot 2\pi$$

which shows the desired inequality.

Proposition 18.2
Let $\Phi : C[-\pi, \pi] \to L^2[-\pi, \pi]$ be the function that sends $f \in C[-\pi, \pi]$ to $\widehat{f}$.
(1) Φ is linear and continuous (so in particular Lipschitz).

(2) Φ is injective.
(3) The range of Φ is dense in $L^2[-\pi,\pi]$.
Proof:
(1) For linearity, this follows by linearity of the quotient map. To show it is Lipschitz, just use the estimate from Remark 18.1 (with Lipschitz constant $\sqrt{2\pi}$).
(2) We show that the nullspace of Φ is $\{0\}$. Taking the contrapositive, we must show that if $f \neq 0$, then $||f||_2 > 0$. Indeed, since $f \neq 0$, there exists $x_0 \in [-\pi,\pi]$ such that $f(x_0) \neq 0$. Moreover, by continuity, letting $|f(x_0)| = \alpha$, we can find $\delta > 0$ such that $(x_0-\delta, x_0+\delta) \subseteq [-\pi,\pi]$ on which $|f| > \frac{\alpha}{2}$. Then, by this estimate, we have $f^2 \geq \frac{\alpha^2}{2}\chi_{(x_0-\delta,x_0+\delta)}$, so integrating we get

$$||f||_2^2 = \int_{[-\pi,\pi]} f^2 d\mu \geq \int_{[-\pi,\pi]} \frac{\alpha^2}{2}\chi_{(x_0-\delta,x_0+\delta)} d\mu = \frac{\alpha^2}{2}2\delta > 0$$

(3) Let $\mathcal{V} = \mathrm{cl}(\mathrm{im}\,\Phi)$. Using Urysohn's Lemma, for a compact set $K \subseteq [-\pi,\pi]$, one shows $\widehat{\chi}_K \in \mathcal{V}$. Then, using the method of friendly sets, one shows that $\widehat{\chi}_B \in \mathcal{V}$ for all $B \in \mathfrak{B}_{[-\pi,\pi]}$ using the previous step. With this, one proves that all equivalence classes of simple functions are in $\mathcal{V}$, and then one proves that the simple functions are dense in $L^2[-\pi,\pi]$ with Fact 15.3. ■

Theorem 18.3
On $L^2[-\pi,\pi]$ it makes sense to define an inner product by putting $\langle \widehat{f}, \widehat{g}\rangle = \langle f, g\rangle = \int_{[-\pi,\pi]} fg d\mu$, which makes $(L^2[-\pi,\pi], \langle\cdot,\cdot\rangle)$ a Hilbert space.

Remark 18.4
In Question 2 of Homework 4, we had an inner product on $C[-\pi,\pi]$, but it was not complete. Thus, Proposition 18.2 and THeorem 18.3 say that $L^2[-\pi,\pi]$ is the completion of this inner product space.

Remark 18.5
The same Question 2 of Homeowrk 4 introduced an orthonormal system $(\xi_n)_{n=1}^\infty$ on $[-\pi,\pi]$ as follows: $\xi_1 \equiv \frac{1}{\sqrt{2\pi}}$, $\xi_{2k}(x) = \frac{1}{\sqrt{\pi}}\sin(kx)$, and $\xi_{2k+1}(x) = \frac{1}{\sqrt{\pi}}\cos(kx)$ for all $k \in \mathbb{N}$ and $x \in [-\pi,\pi]$. It can be shown (withe Stone-Weierstrass and Remark 18.1) that $(\widehat{\xi}_n)_{n=1}^\infty$ is an orthonormal basis of $L^2[-\pi,\pi]$.

Definition 18.6
For every $f \in \mathcal{L}^2[-\pi,\pi]$, the ξ-coefficients of $\widehat{f} \in L^2[-\pi,\pi]$ with respect to the above orthonormal basis $(\widehat{\xi}_n)_{n=1}^\infty$, given by $(c_n)_{n=1}^\infty = (\langle f, \xi_n\rangle)_{n=1}^\infty$ are called the Fourier coefficients of f.

Remark 18.7
The machinery of Lecture 5 applies, in particular the Riesz-Fischer theorem (Theorem 5.4) and Parseval's formula (Corollary 5.6).

Example 18.8
Let $f : [-\pi,\pi]$ be given by $f(t) = t$. We compute its Fourier coefficients:

$$c_1 = \int_{-\pi}^{\pi} = \frac{1}{\sqrt{2\pi}} t dt = 0$$

and in general, every odd Fourier coefficient is zero. For the even coefficients, by integration by parts,

$$c_{2k} = \int_{-\pi}^{\pi} \frac{1}{\sqrt{\pi}}\sin(kt) t dt = \frac{(-1)^{k+1}2\sqrt{\pi}}{k}$$

Then Parseval's formula says

$$\frac{2\pi^3}{3} = \int_{-\pi}^{\pi} t^2 dt = ||f||_2^2 = \sum_{k=1}^{\infty} \frac{4\pi}{k^2}$$

so simplifying we get

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$$

This completes the course.